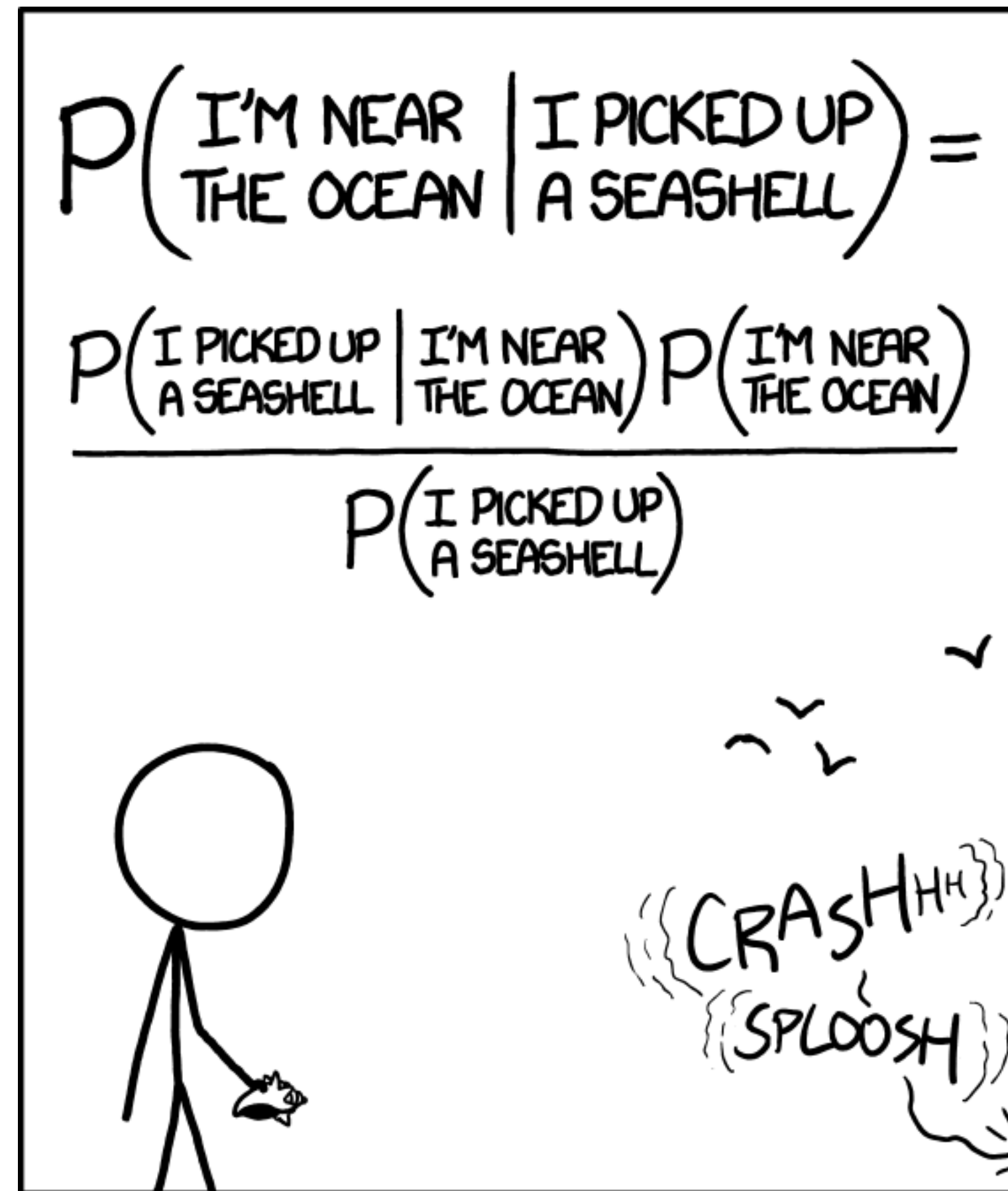
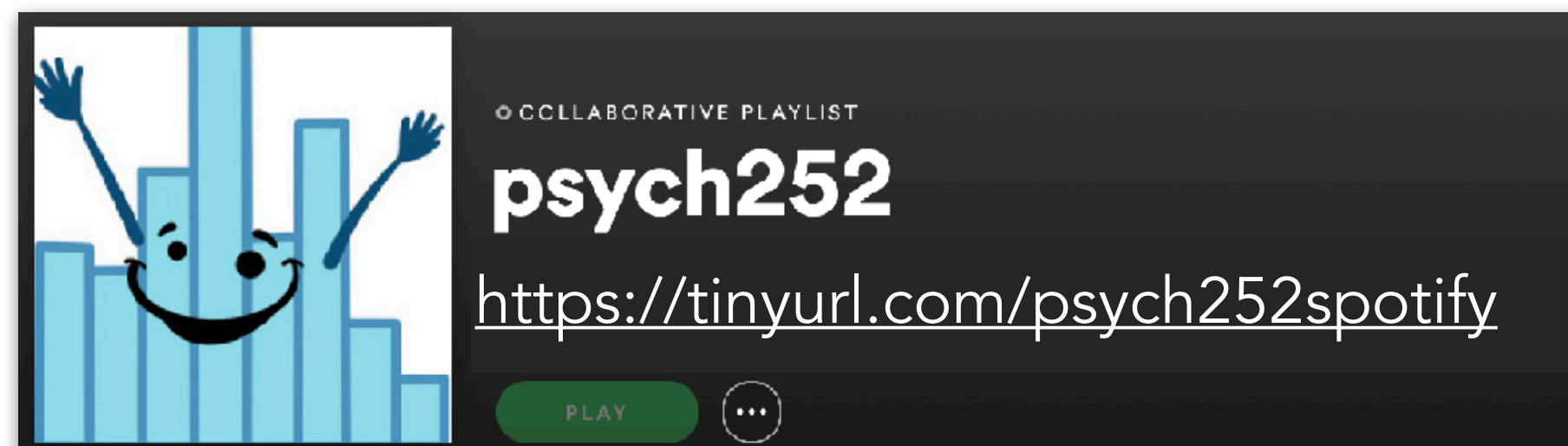


Bayesian data analysis 4



STATISTICALLY SPEAKING, IF YOU PICK UP A SEASHELL AND *DON'T* HOLD IT TO YOUR EAR, YOU CAN PROBABLY HEAR THE OCEAN.



03/06/2026

Logistics

Final presentation survey → goal: everyone!

Questions Responses **21** Settings

Final presentation

Thanks for filling out this survey to help us with planning!

How are you planning to present? *

☐ In class (preferred option if possible)

☐ Remotely (live)

☐ I will record the presentation and submit a video before March 15th.

☐ Other...

What's your name (e.g. Tobias Gerstenberg)? *

Short answer text

What's the name of your team's github repository (e.g. final-project-tobi)? *

Short answer text

How many people are in your team (e.g. 1, 2, or 3)?

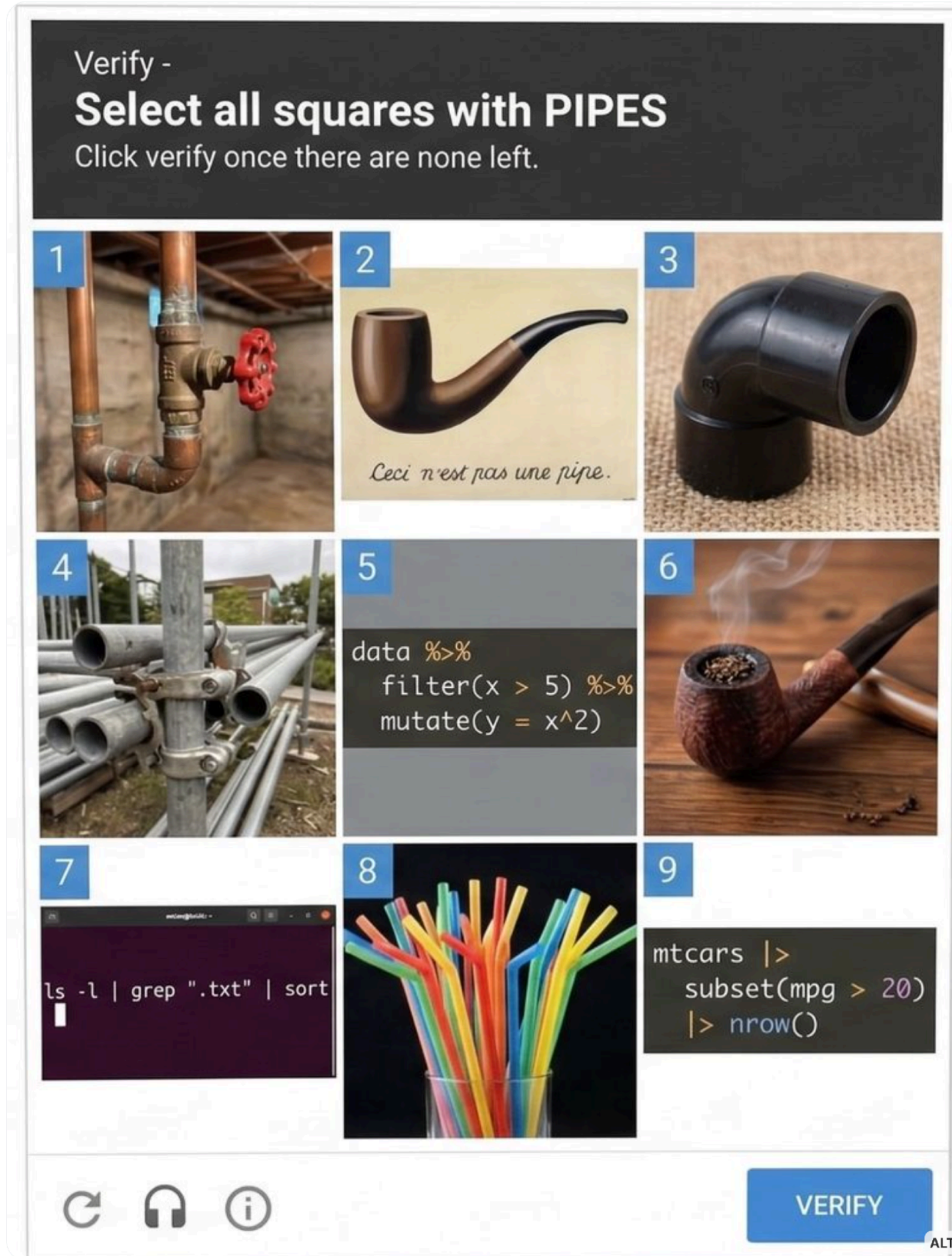
Short answer text

<https://tinyurl.com/psych252presentation26>

Things that came up

 **Hadley Wickham**
@hadley.nz

These captchas just keep getting harder [#rstats](#)



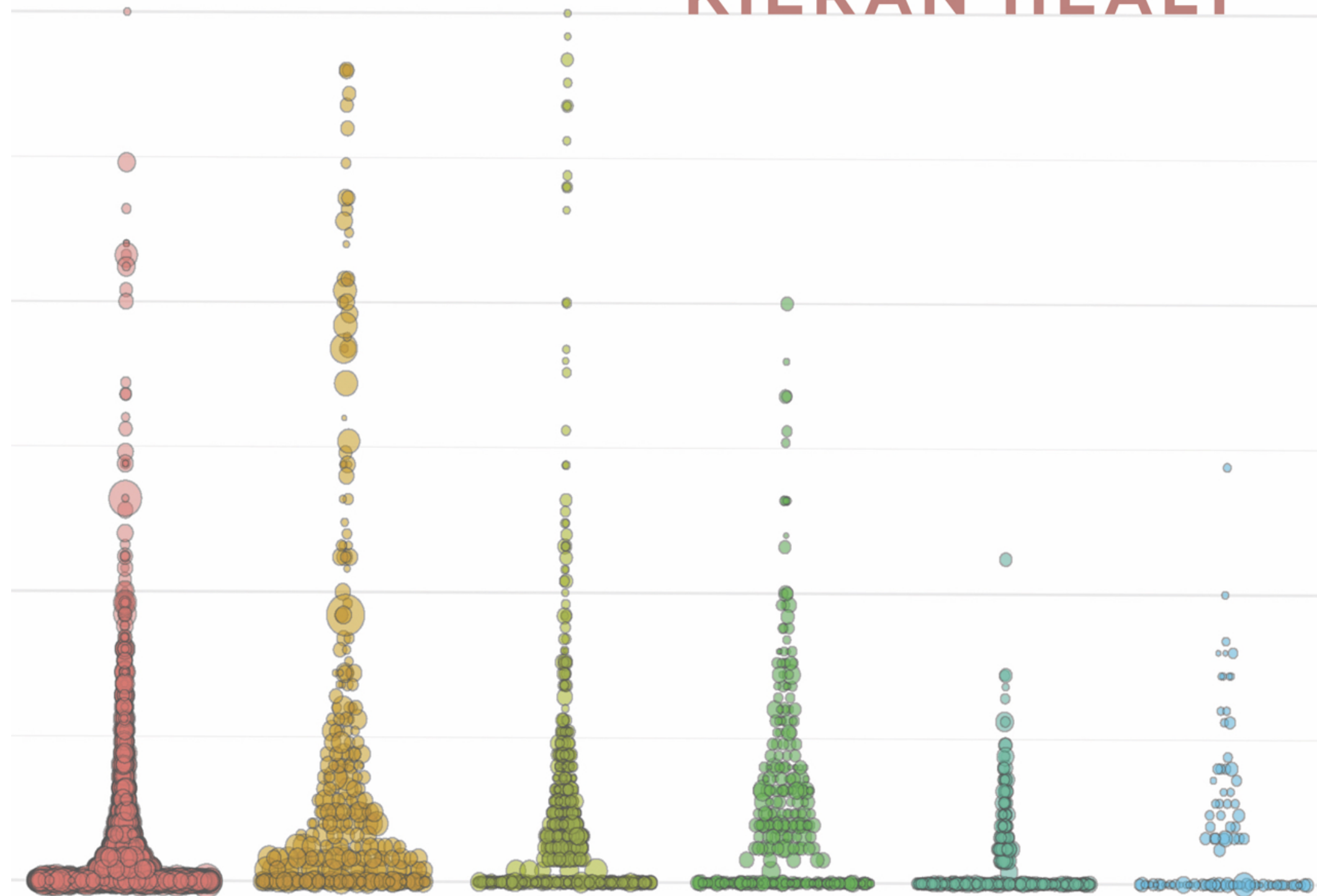
7:58 AM · Mar 5, 2026  Everybody can reply

122 reposts 2 quotes 517 likes 9 saves

DATA VISUALIZATION

A PRACTICAL INTRODUCTION

KIERAN HEALY



Preface

- 1 Look at Data
 - 2 Get Started
 - 3 Make a Plot
 - 4 Show the Right Numbers
 - 5 Graph Tables, Add Labels, Make Notes
 - 6 Work with Models
 - 7 Draw Maps
 - 8 Refine your Plots
- Appendix
Acknowledgments

Data Visualization

A Practical Introduction

Second Edition

Kieran Healy

March 2026

Note

This is the website for the second edition of *Data Visualization: A Practical Introduction*, forthcoming from [Princeton University Press](#). It contains a complete draft of the manuscript. The book is not yet available for pre-order. But, if you would like to receive **one** (and only one) email to let you know when you can order it, please [fill out this form](#). Your email will be used for this single purpose only and will not be shared with anyone.

In this Chapter

Preface

- What's New in This Edition?
- What You Will Learn
- The Right Frame of Mind
- How to Use This Book
- Conventions
- Before You Begin

<https://socviz.co/>

check out the book!

Plan for today

- Quick recap
- Doing Bayesian data analysis **with BRMS**
 - Reporting results
- Some more examples
 - Sleep data
 - Titanic data
- Going beyond
 - Evidence for the null hypothesis
 - I want only positive coefficients
 - Dealing with unequal variance

Quick recap



XX

Quick recap: Doing Bayesian data analysis

Software packages



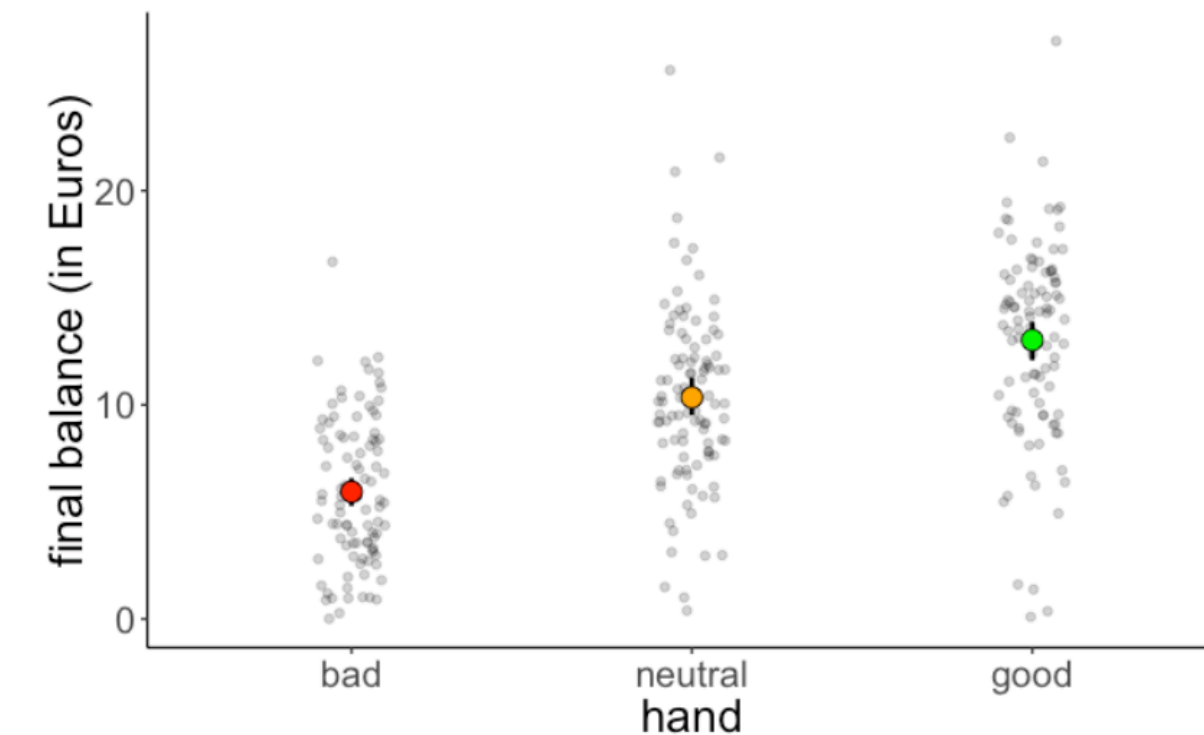
Bayesian regression
modeling with Stan

`library("brms")`

- very powerful package that makes it easy to run Bayesian regression models
- we specify models using the same syntax we've already learned based on `lm()`, `glm()`, and `lmer()`
- `brms` turns this into Stan code and fits the model
- we can then use `tidybayes` to investigate the posterior

34

Poker data



37

Using `brm()`

```
1 fit_brm_poker = brm(formula = balance ~ 1 + hand,  
2 data = df.poker)  
3  
4 fit_brm_poker %>% summary()
```

cool!

```
Family: gaussian  
Links: mu = identity; sigma = identity  
Formula: balance ~ 1 + hand  
Data: df.poker (Number of observations: 300)  
Samples: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;  
total post-warmup samples = 4000  
  
Population-Level Effects:  
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS  
Intercept      5.93      0.41    5.12    6.72 1.00    2986    2744  
handneutral     4.41      0.58    3.30    5.55 1.00    3497    2903  
handgood        7.10      0.58    5.99    8.29 1.00    3545    2932  
  
Family Specific Parameters:  
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS  
sigma      4.12      0.17    3.81    4.46 1.00     3650     2921  
  
Samples were drawn using sampling(NUTS). For each parameter, Eff.Sample  
is a crude measure of effective sample size, and Rhat is the potential  
scale reduction factor on split chains (at convergence, Rhat = 1).
```

39

"Full" specification of the model

```
1 fit_brm_poker_full = brm(  
2 formula = balance ~ 1 + hand,  
3 family = "gaussian",  
4 data = df.poker,  
5 prior = c(  
6   prior(normal(0, 10), class = "b", coef = "handgood"),  
7   prior(normal(0, 10), class = "b", coef = "handneutral"),  
8   prior(student_t(3, 3, 10), class = "Intercept"),  
9   prior(student_t(3, 0, 10), class = "sigma")  
10 ),  
11 inits = list(  
12   list(Intercept = 0, sigma = 1, handgood = 5, handneutral = 5),  
13   list(Intercept = -5, sigma = 3, handgood = 2, handneutral = 2),  
14   list(Intercept = 2, sigma = 1, handgood = -1, handneutral = 1),  
15   list(Intercept = 1, sigma = 2, handgood = 2, handneutral = -2)  
16 ),  
17 iter = 4000,  
18 warmup = 1000,  
19 chains = 4,  
20 file = "cache/brm_poker_full",  
21 seed = 1  
22 )
```

likelihood

priors

initialization

how many runs in the inference chain

how long for the warmup

how many chains

save the model result

make reproducible

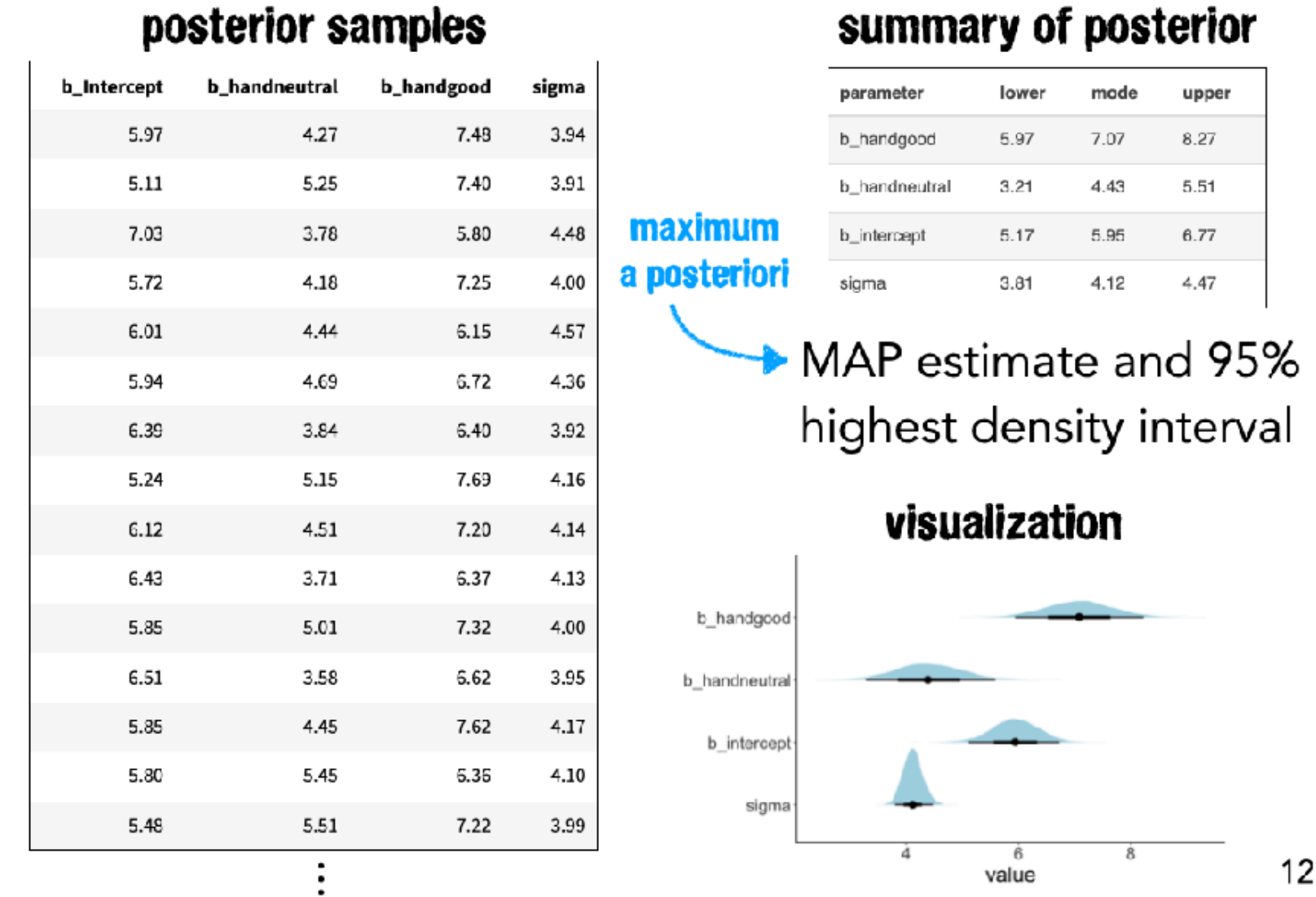
fitting Bayesian models takes some time, so storing results is key

42

10

Quick recap: Testing hypotheses

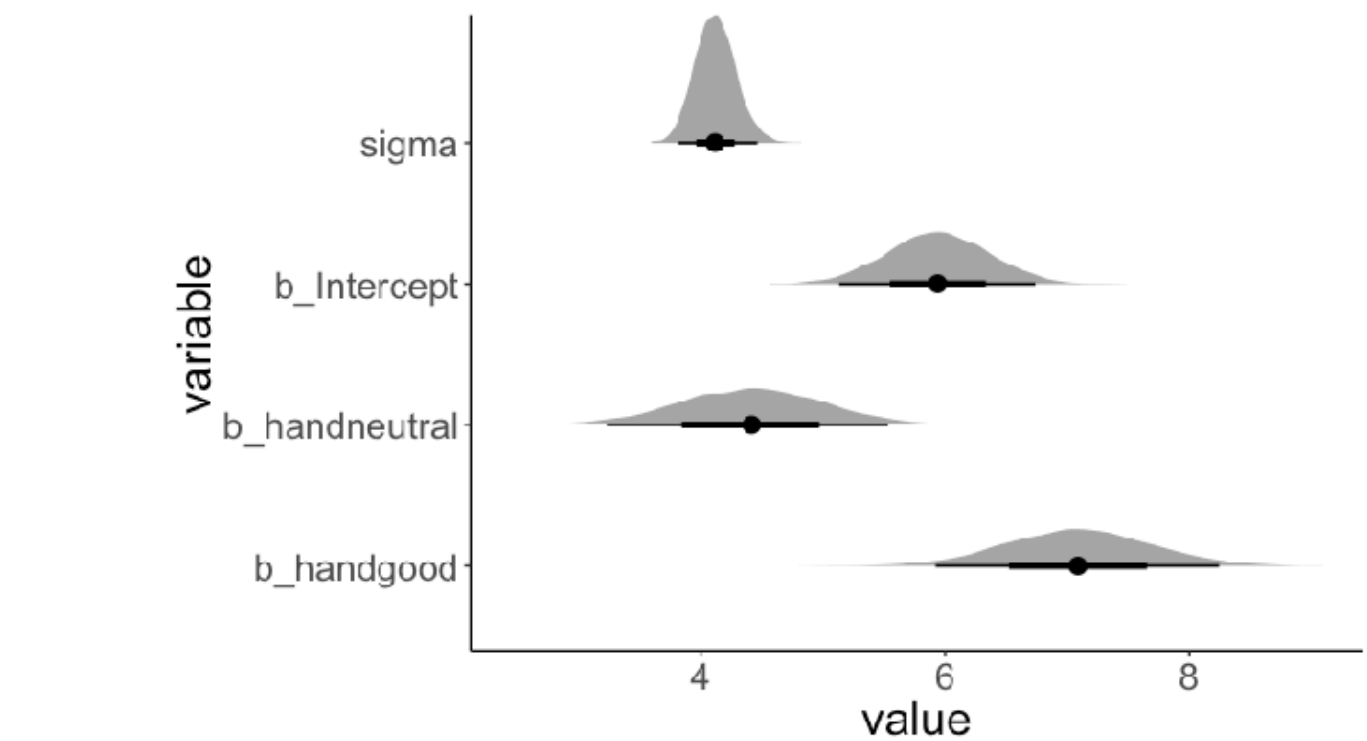
Results



12

Asking questions based on the posterior

Do good hands make twice as much as bad hands?



```
1 hypothesis(fit_brm,  
2           hypothesis = "handgood + Intercept > 2 * Intercept")
```

p = 0.89

16

Testing hypothesis

```
1 df.hypothesis = fit_brm %>%  
2   posterior_samples() %>%  
3   clean_names() %>%  
4   select(starts_with("b_")) %>%  
5   mutate(neutral = b_intercept + b_handneutral,  
6          bad_good_average = (b_intercept + b_intercept + b_handgood)/2,  
7          hypothesis = neutral < bad_good_average)
```

samples from the posterior

b_intercept	b_handneutral	b_handgood	neutral	bad_good_average	hypothesis
6.07	4.10	7.20	10.17	9.67	FALSE
6.06	4.44	6.95	10.49	9.53	FALSE
5.88	5.00	8.73	10.87	9.24	FALSE
5.85	4.78	8.16	10.63	8.94	FALSE
5.86	4.46	7.68	10.32	9.70	FALSE

```
1 df.hypothesis %>%  
2   summarize(p = sum(hypothesis)/n())
```

p = 0.04

18

The "emmeans" package is your friend!

```
1 fit_brm_poker %>%  
2   emmeans(specs = consec ~ hand)
```

estimated mean for each group

hand	emmean	lower.HPD	upper.HPD
bad	5.94	5.16	6.78
neutral	10.34	9.55	11.15
good	13.02	12.22	13.82

Point estimate displayed: median
HPD interval probability: 0.95

contrast	estimate	lower.HPD	upper.HPD
neutral - bad	4.38	3.24	5.52
good - neutral	2.69	1.51	3.78

Point estimate displayed: median
HPD interval probability: 0.95

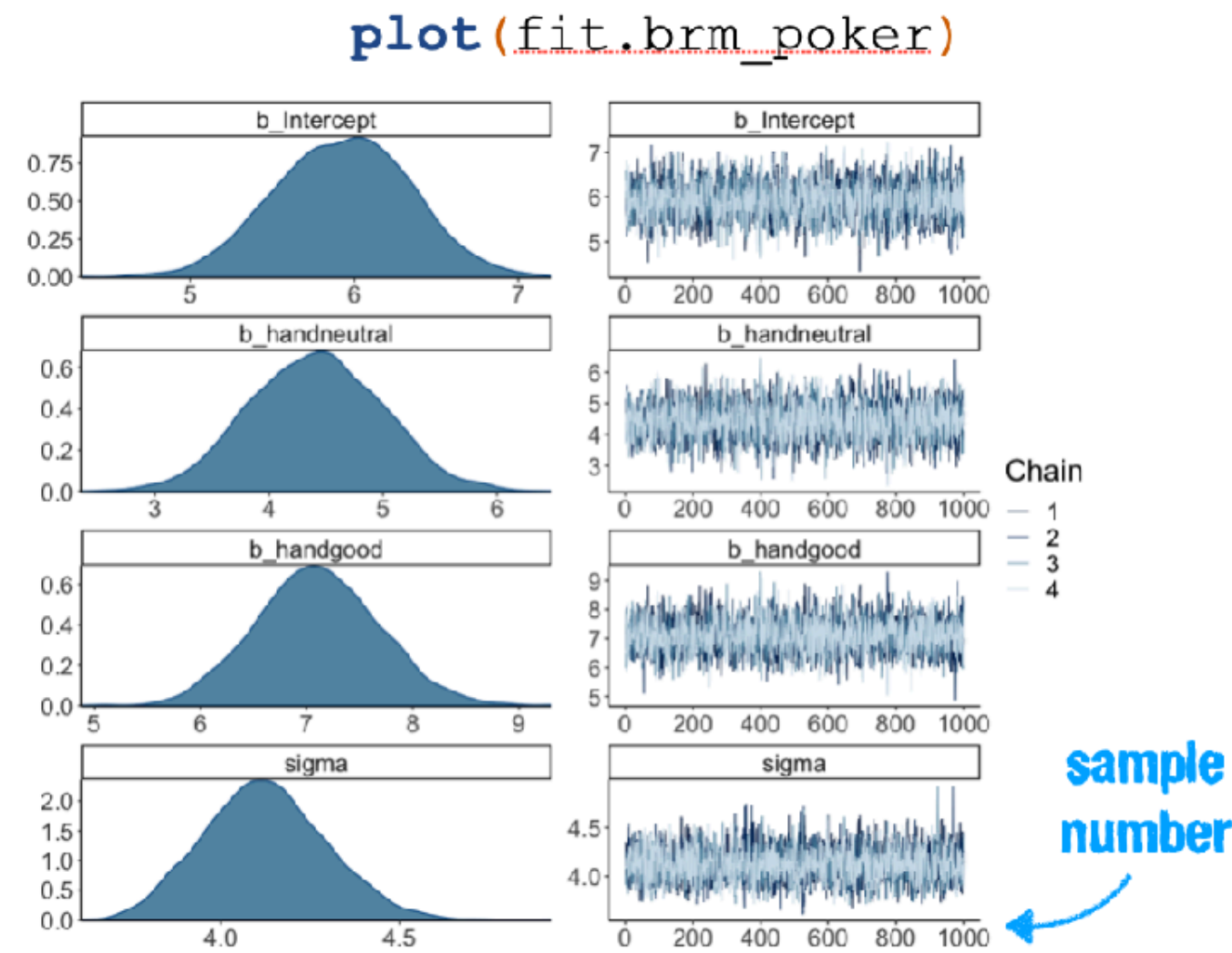
contrasts

20

11

Quick recap: Model evaluation

Can we trust the model results?

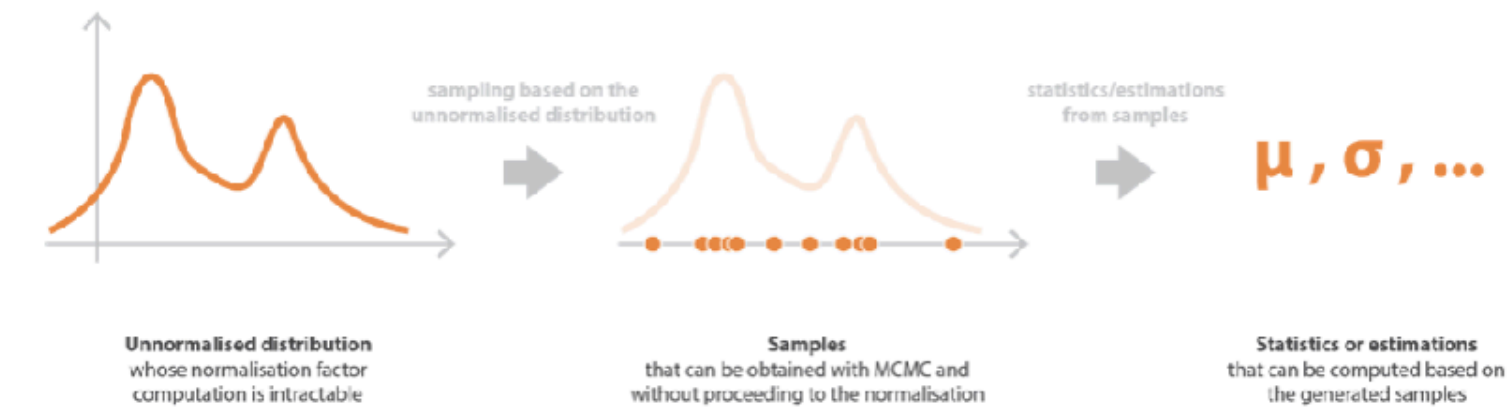


<https://chi-feng.github.io/mcmc-demo/>

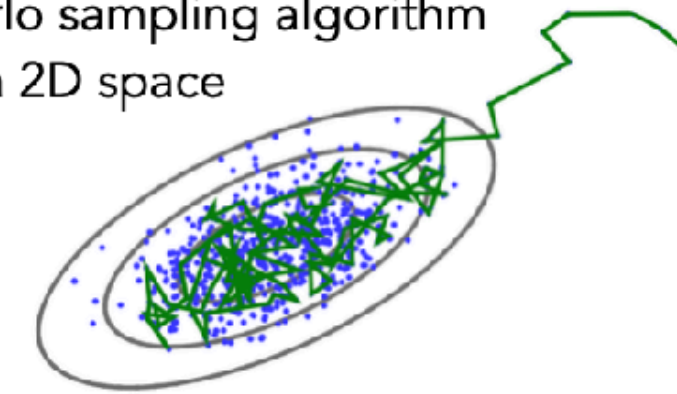
24

Can we trust the model results?

Inference via Markov Chain Monte Carlo (MCMC)



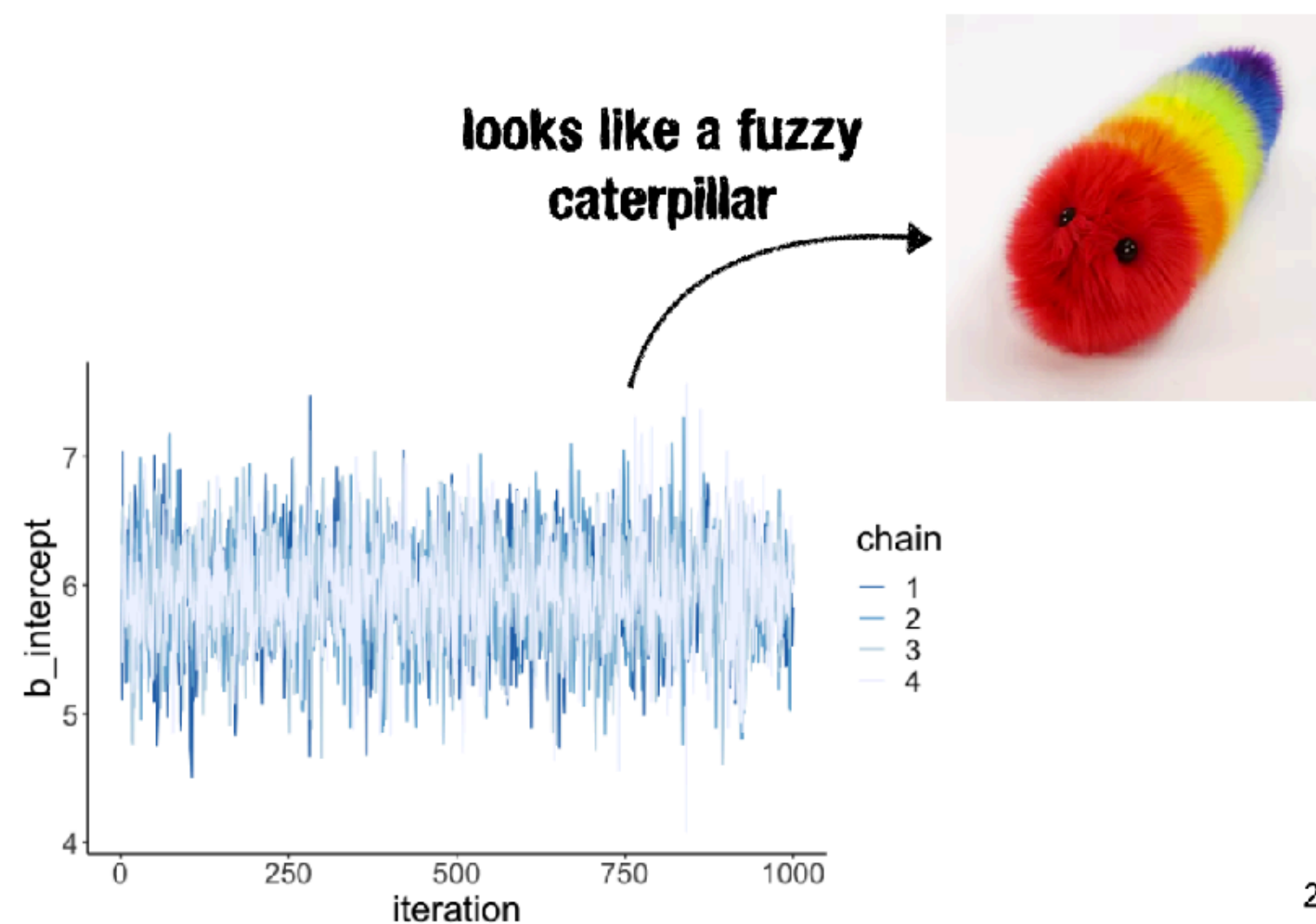
Markov Chain Monte Carlo sampling algorithm in a 2D space



goal: draw independent samples from the posterior distribution

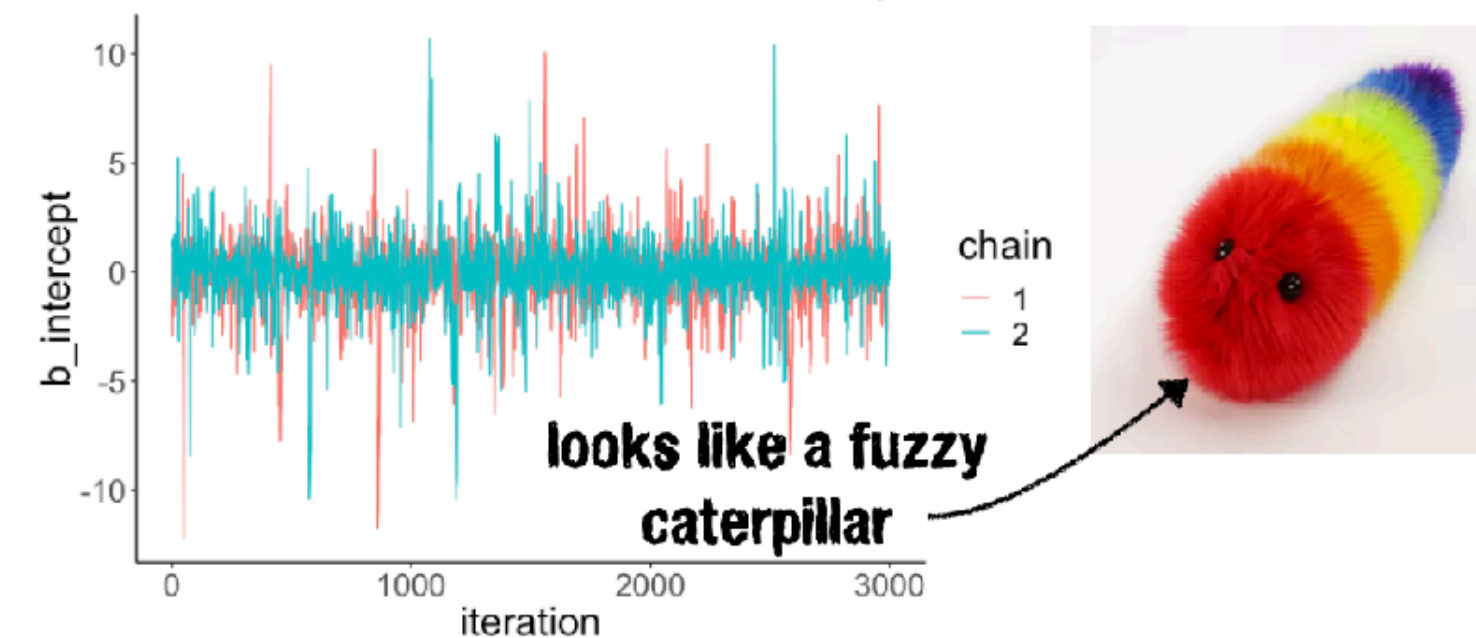
25

Can we trust the model results?



26

Having somewhat informative priors fixes things



if things go wrong:

- set more informative priors
- run more warm-up samples
- adjust the sampling algorithm as suggested via the control argument

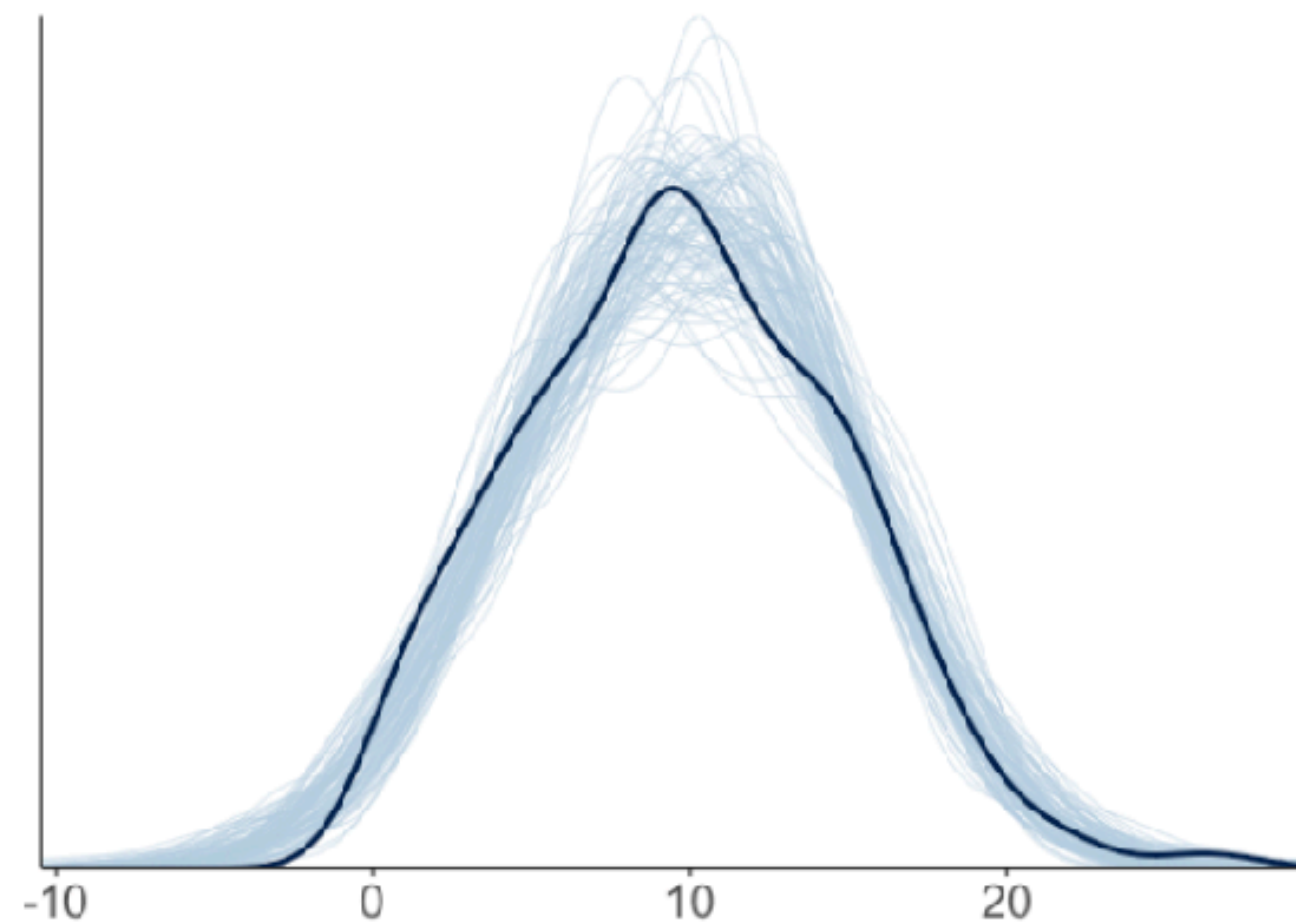
32

12

Quick recap: Posterior predictive check & reporting

Posterior predictive check

```
pp_check(fit.brm, nsamples = 100)
```



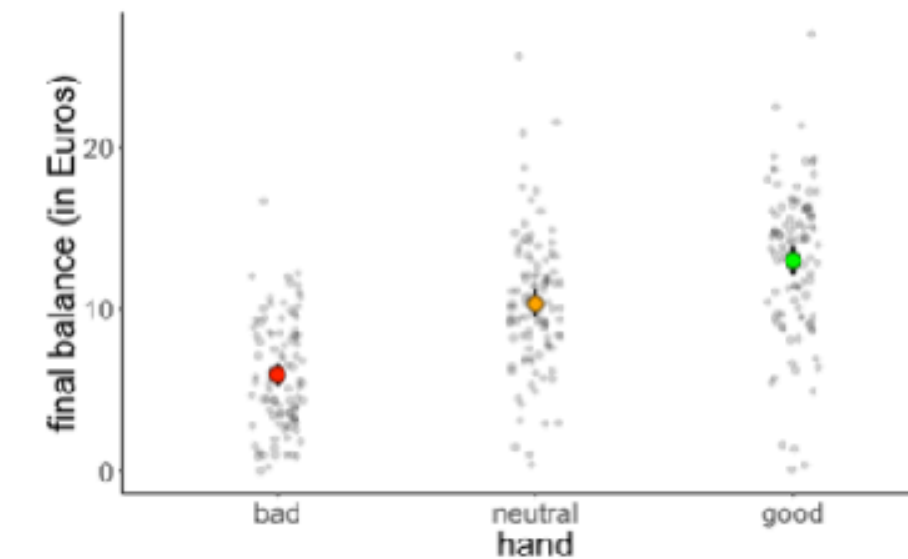
— y
— y_{rep}

The model accurately captures the distribution of the response variable

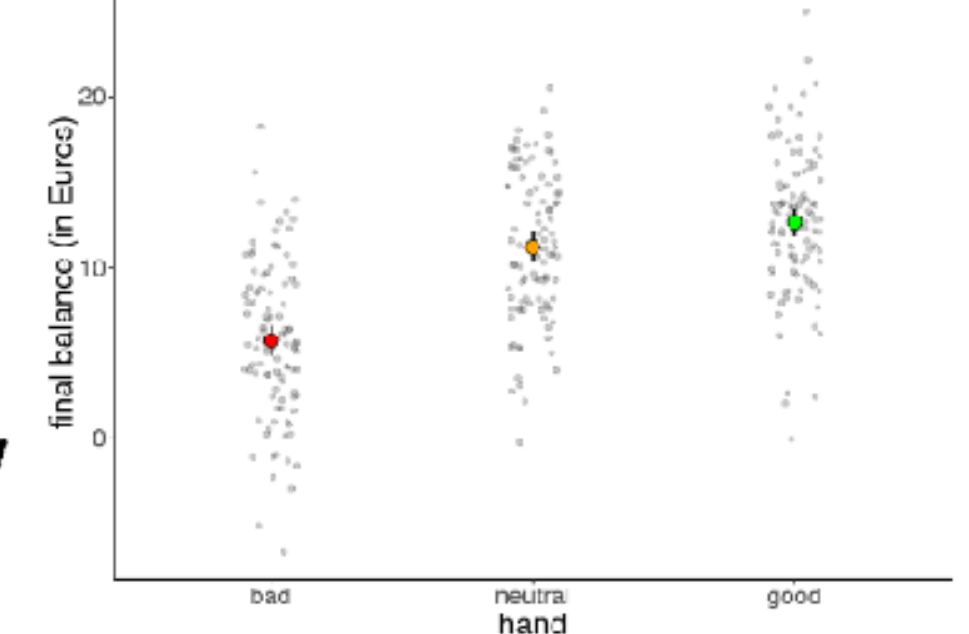
34

Posterior predictive check

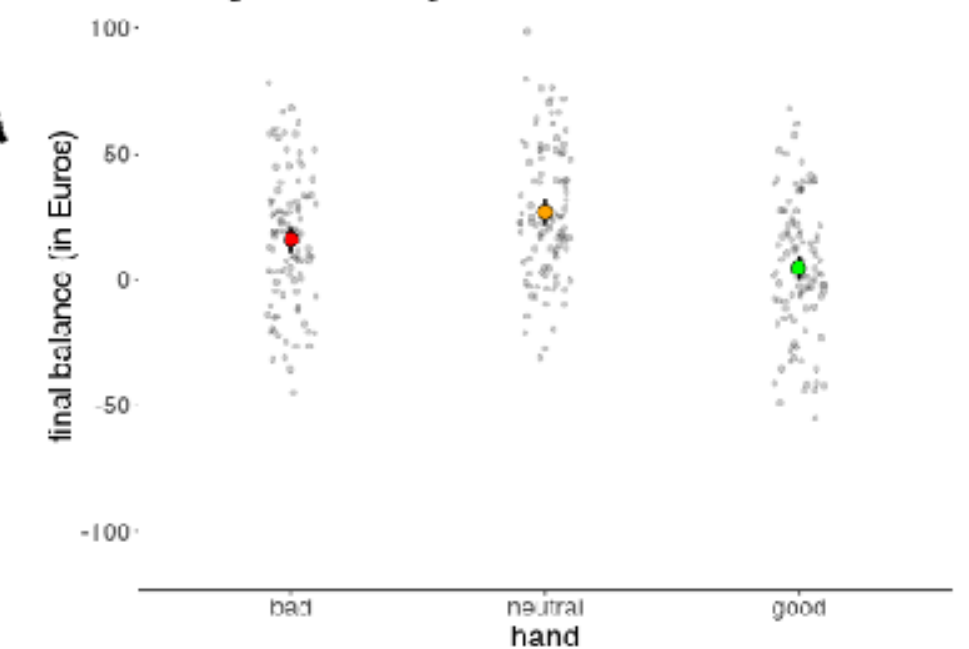
original data



posterior prediction



prior prediction



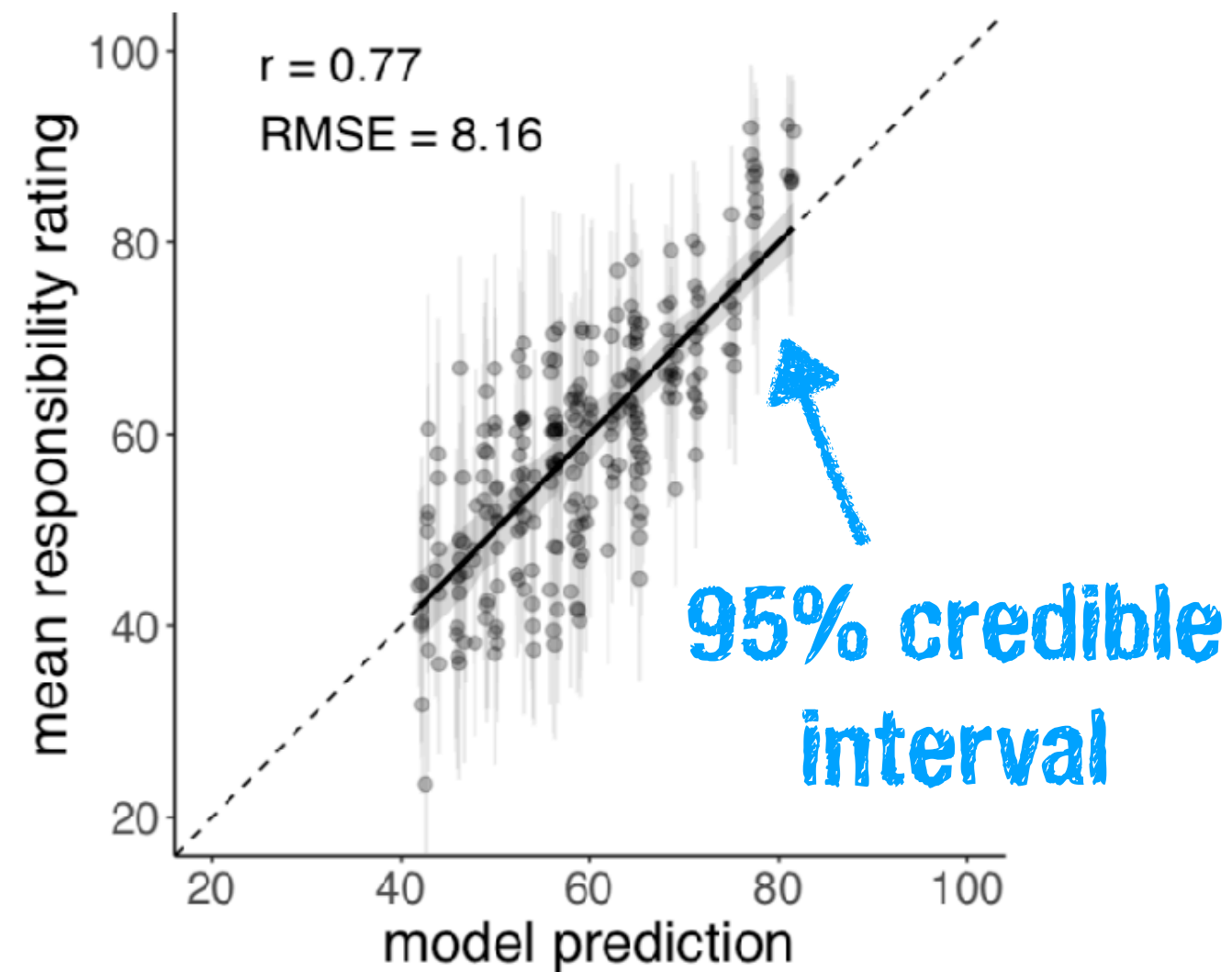
**take a look at course notes
for how to make these**

35

Reporting results

Reporting results

Plots



Tables

Table 1
Estimates of the mean, standard error, and 95% HDIs of the different predictors in the Bayesian mixed effects model. Note: `n_causes` = number of causes.

responsibility $\sim$ 1 + surprise + pivotality + n_causes + (1 + surprise + pivotality + n_causes | participant)

term	estimate	std.error	lower 95% HDI	upper 95% HDI
intercept	59.94	3.25	54.70	65.22
surprise	21.68	4.57	14.17	29.23
pivotality	13.52	1.82	10.47	16.53
n_causes	-5.72	0.50	-6.55	-4.90

model
formula

parameter
estimates

can also include
Random Effects

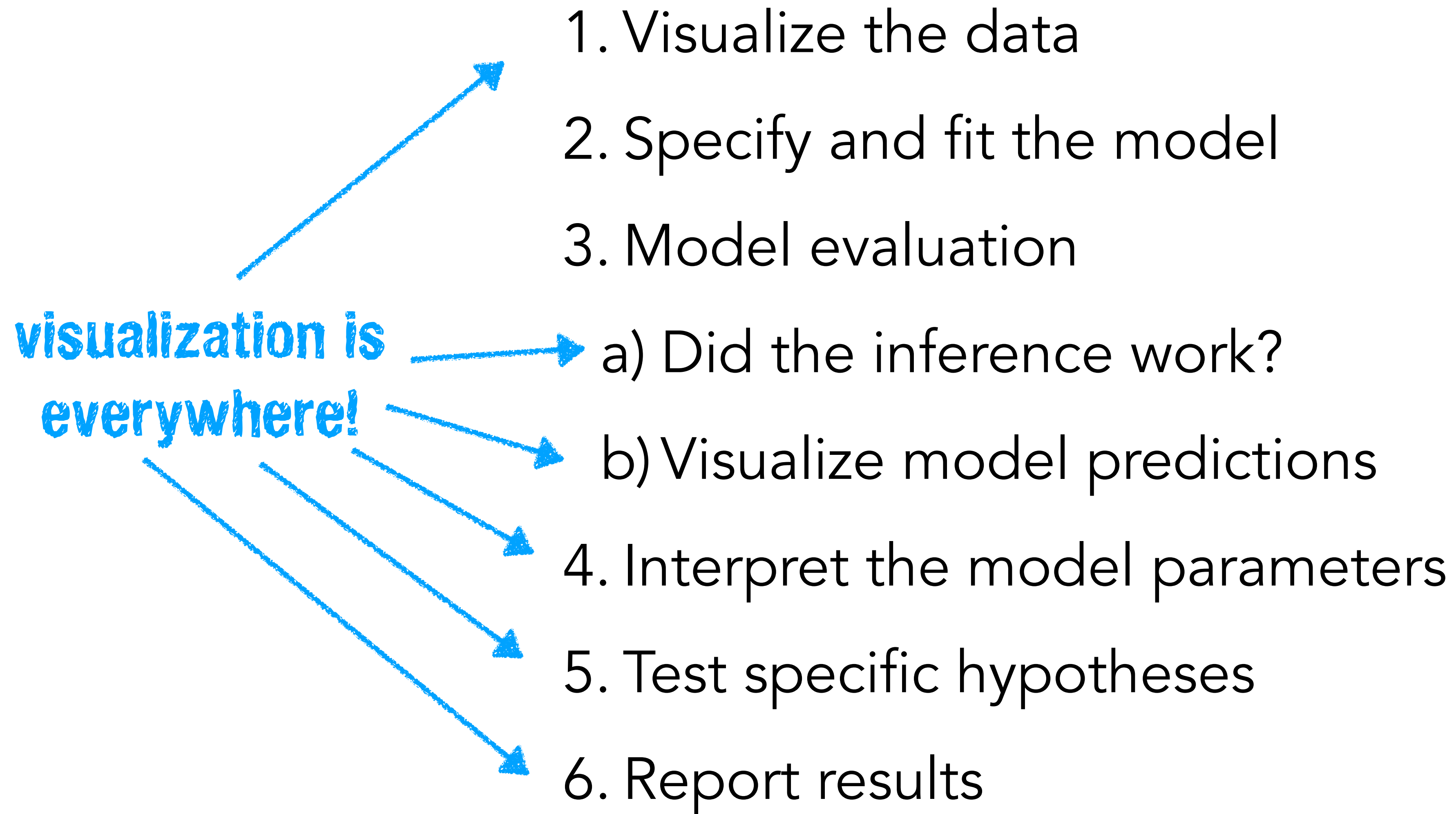
Text

We computed a Bayesian mixed effects model with random intercepts and slopes to predict participants' responsibility judgments (see Table 1). Figure 6b shows a scatter plot of the model predictions and participants' responsibility judgments for the full set of 170 scenarios (with 250 judgments). Overall, the model predicts participants' responsibility judgments well with $r = .77$ and $\text{RMSE} = 8.16$. Table 1 shows the estimates of the different predictors. As can be seen, none of the predictors' 95% HDIs overlap with 0.¹

¹For any statistical claim, we report the mean of the posterior distribution together with the 95% highest-density interval (HDI). All Bayesian models were written in Stan (Carpenter et al., 2017) and accessed with the brms package (Bürkner, 2017) in R (R Core Team, 2019).

Some more examples

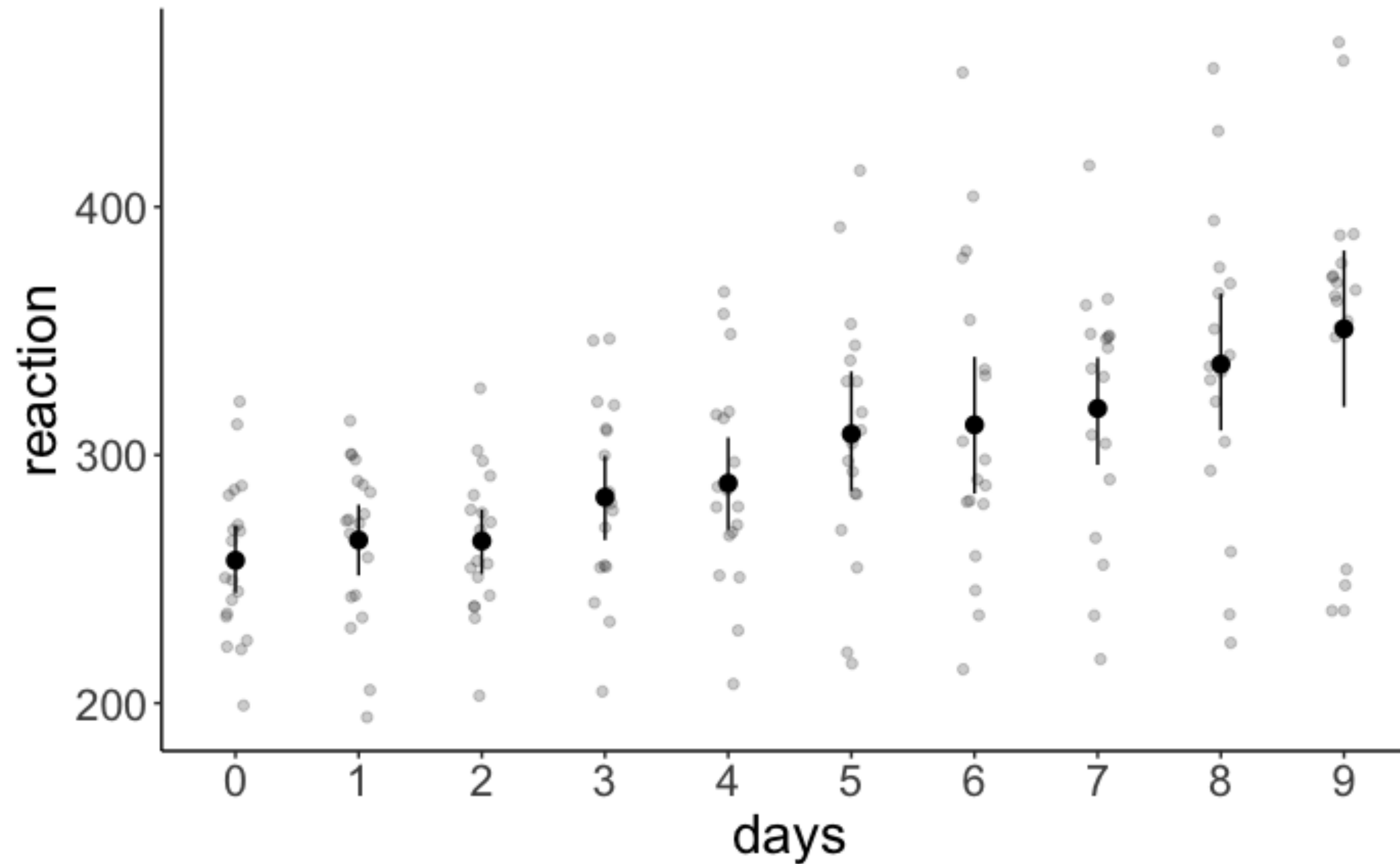
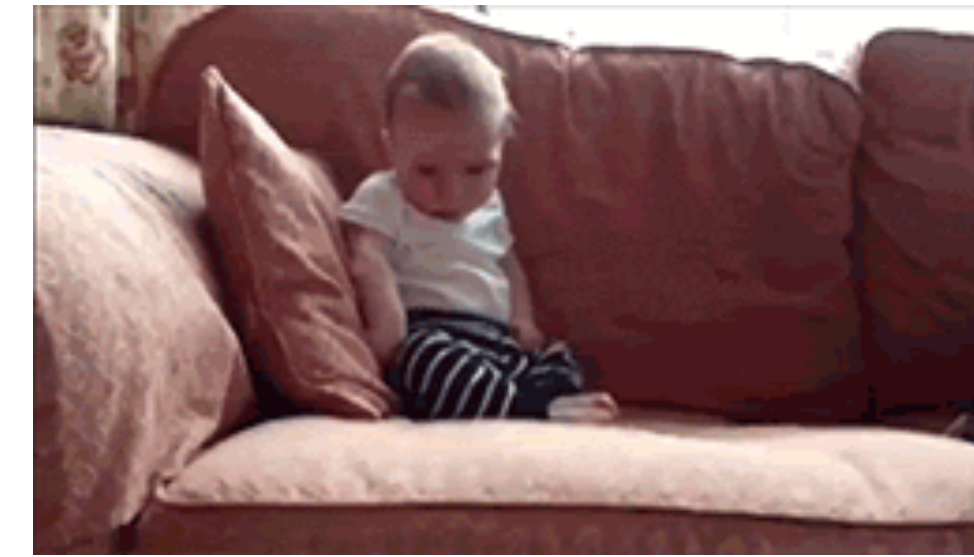
Recipe for Bayesian analysis with brms



Sleep data

1. Visualize the data

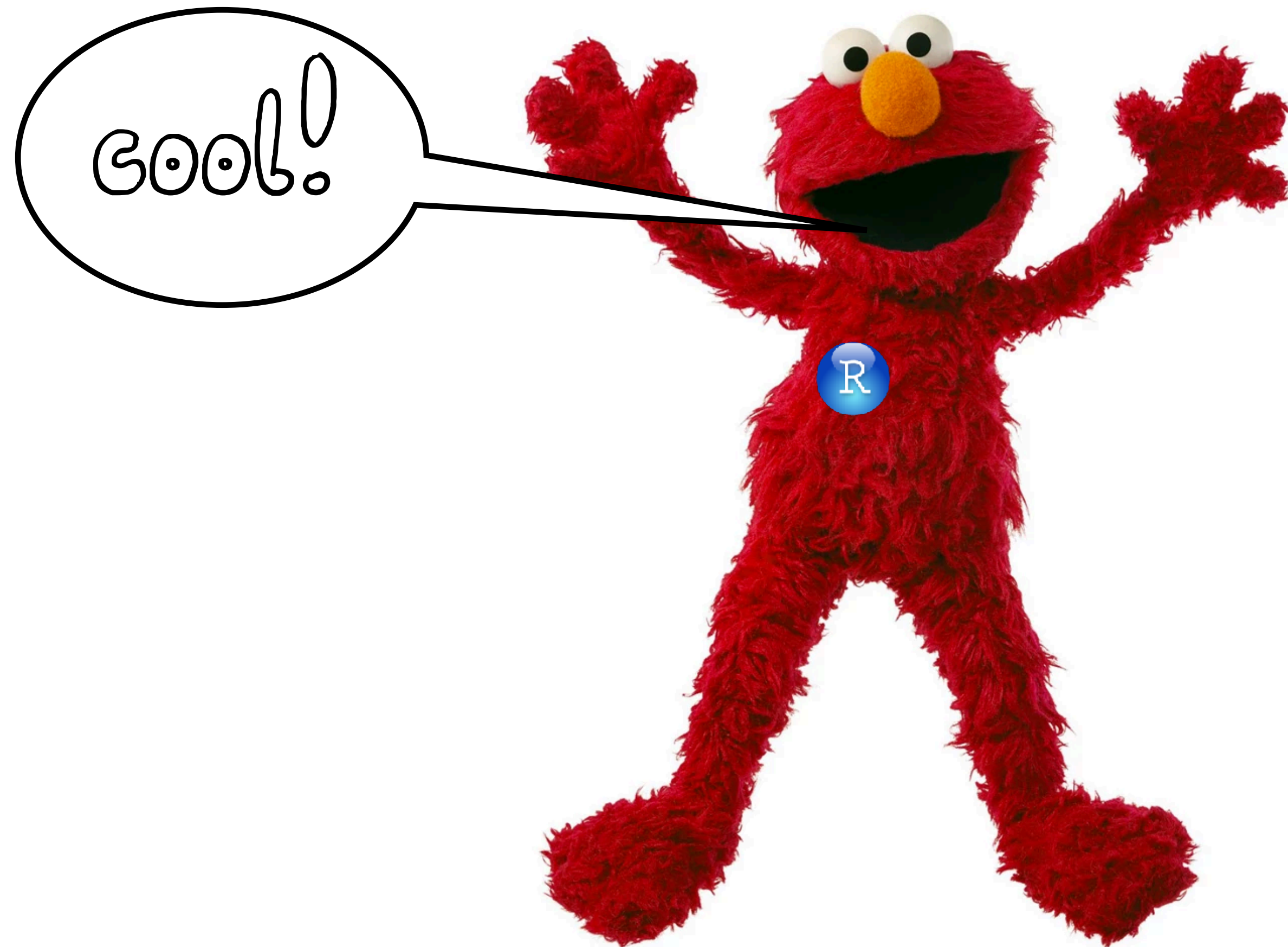
Feeling sleepy?



2. Specify and fit the model

1. Specify and fit the model

```
1 fit.brm_sleep = brm(formula = reaction ~ 1 + days + (1 + days | subject),  
2                       data = df.sleep,  
3                       seed = 1,  
4                       file = "cache/brm_sleep")
```



3. Model evaluation

a) Did the inference work?

```
1 fit.brm_sleep %>%  
2 summary()
```

Family: gaussian
Links: mu = identity; sigma = identity
Formula: reaction ~ 1 + days + (1 + days | subject)
Data: df.sleep (Number of observations: 183)
Samples: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
total post-warmup samples = 4000

Group-Level Effects:
~subject (Number of levels: 20)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	26.18	6.25	15.65	40.54	1.00	1879	2463
sd(days)	6.59	1.53	4.14	10.13	1.00	1145	1625
cor(Intercept,days)	0.09	0.29	-0.46	0.67	1.00	993	1526

Population-Level Effects:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	252.18	6.86	238.47	265.42	1.00	1826	2766
days	10.46	1.69	7.13	13.78	1.00	1203	1782

Family Specific Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	25.77	1.57	22.93	29.14	1.00	3864	2773

Samples were drawn using sampling(NUTS). For each parameter, Eff.Sample is a crude measure of effective sample size, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

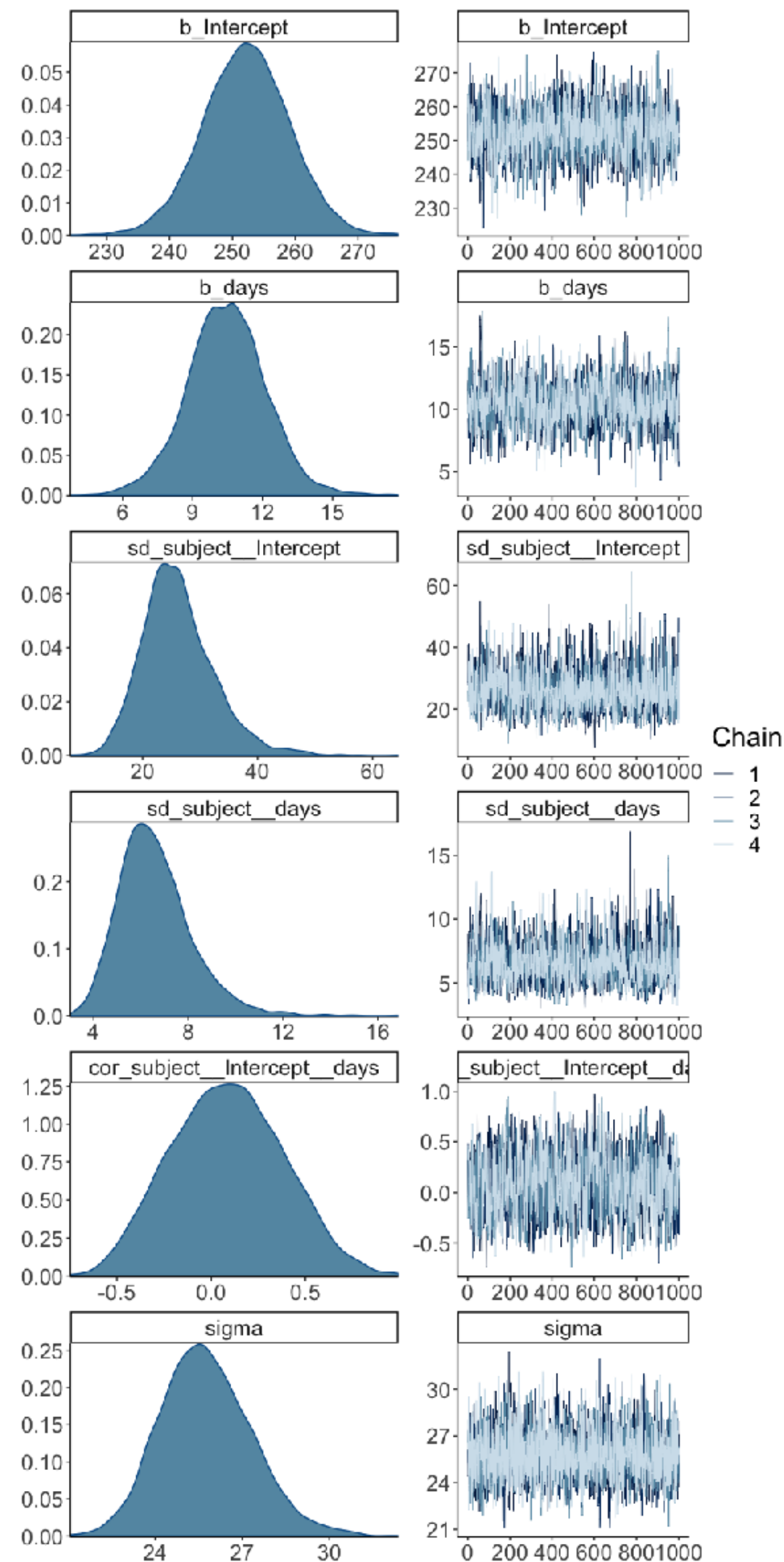
Rhat of 1.00 is good!

Roughly speaking, the effective sample size (**ESS**) of a quantity of interest captures how many independent draws contain the same amount of information as the dependent sample obtained by the MCMC algorithm.

a) Did the inference work?

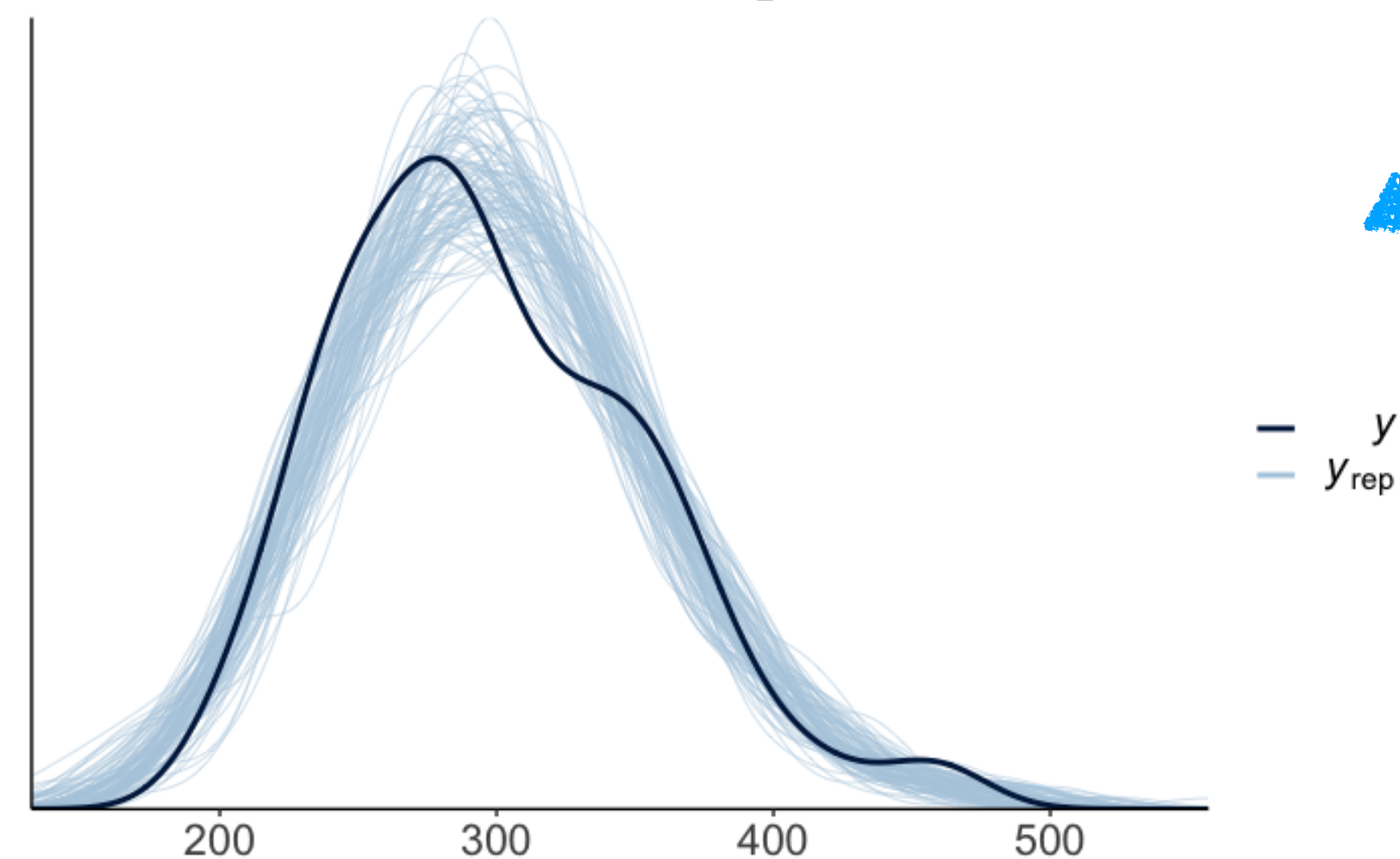
```
1 fit.brm_sleep %>%  
2   plot(N = 6)
```

these look good!

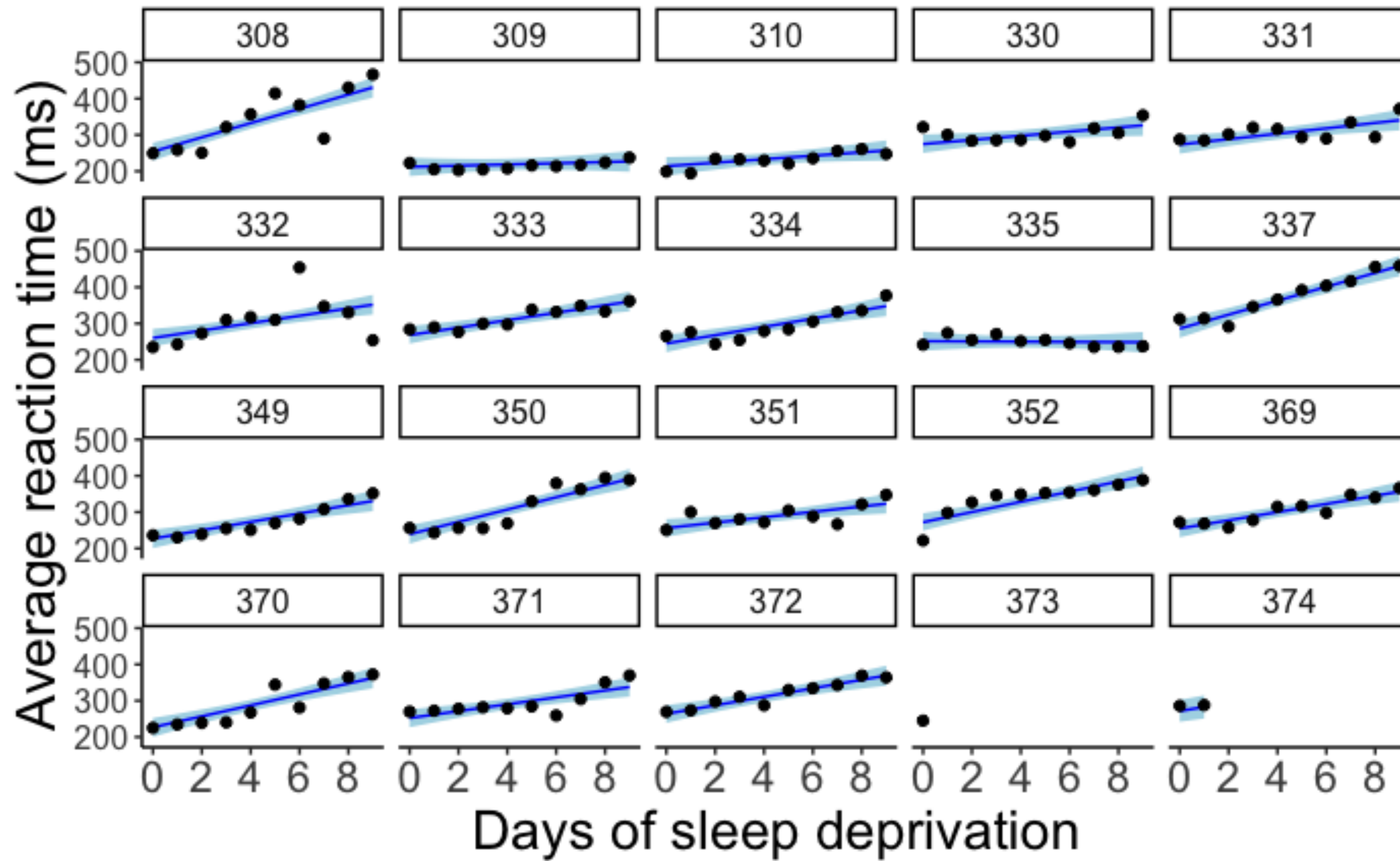


also looks good!

```
1 pp_check(fit.brm_sleep,  
2          nsamples = 100)
```

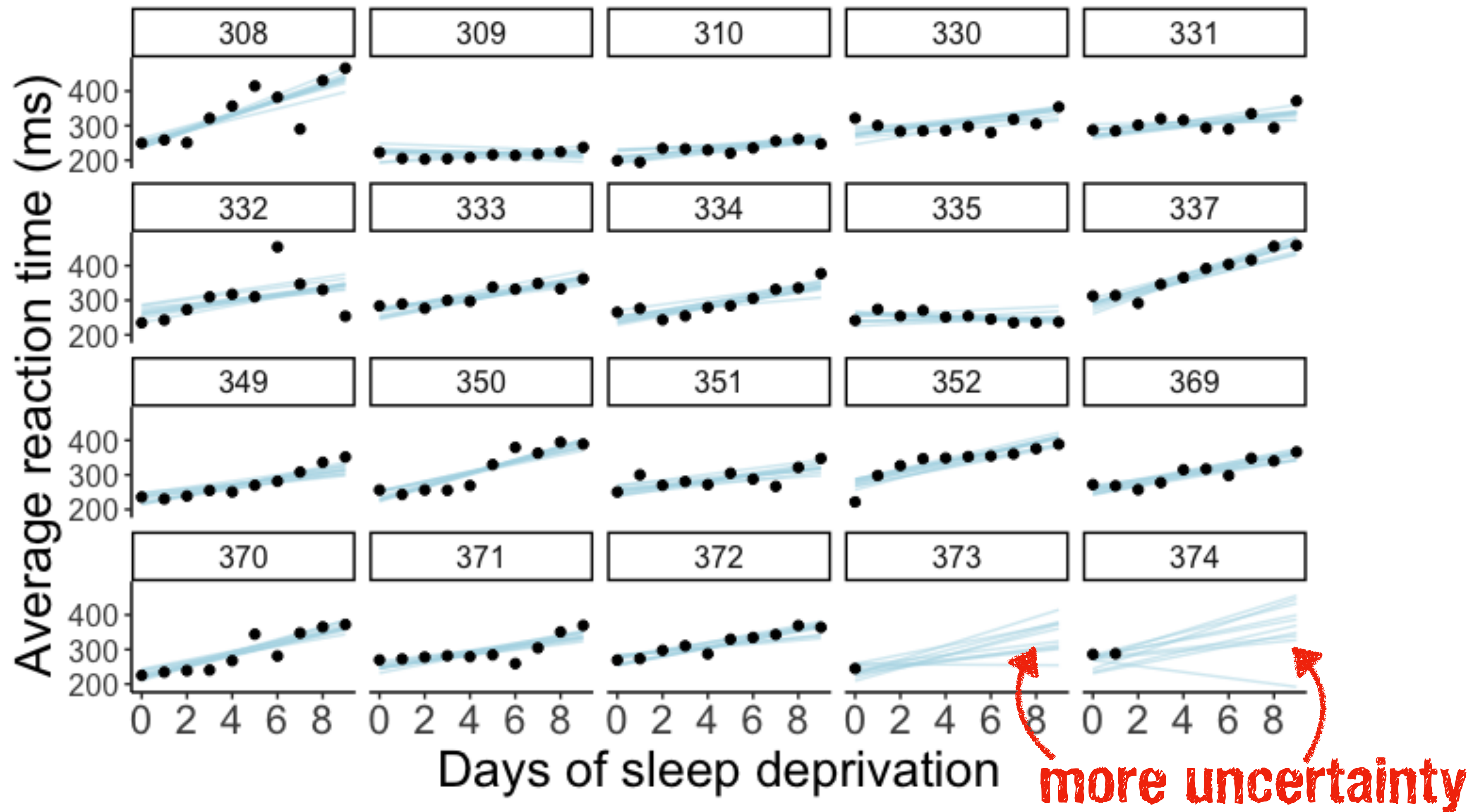


b) Visualize the model predictions



regression lines with 95% highest density intervals

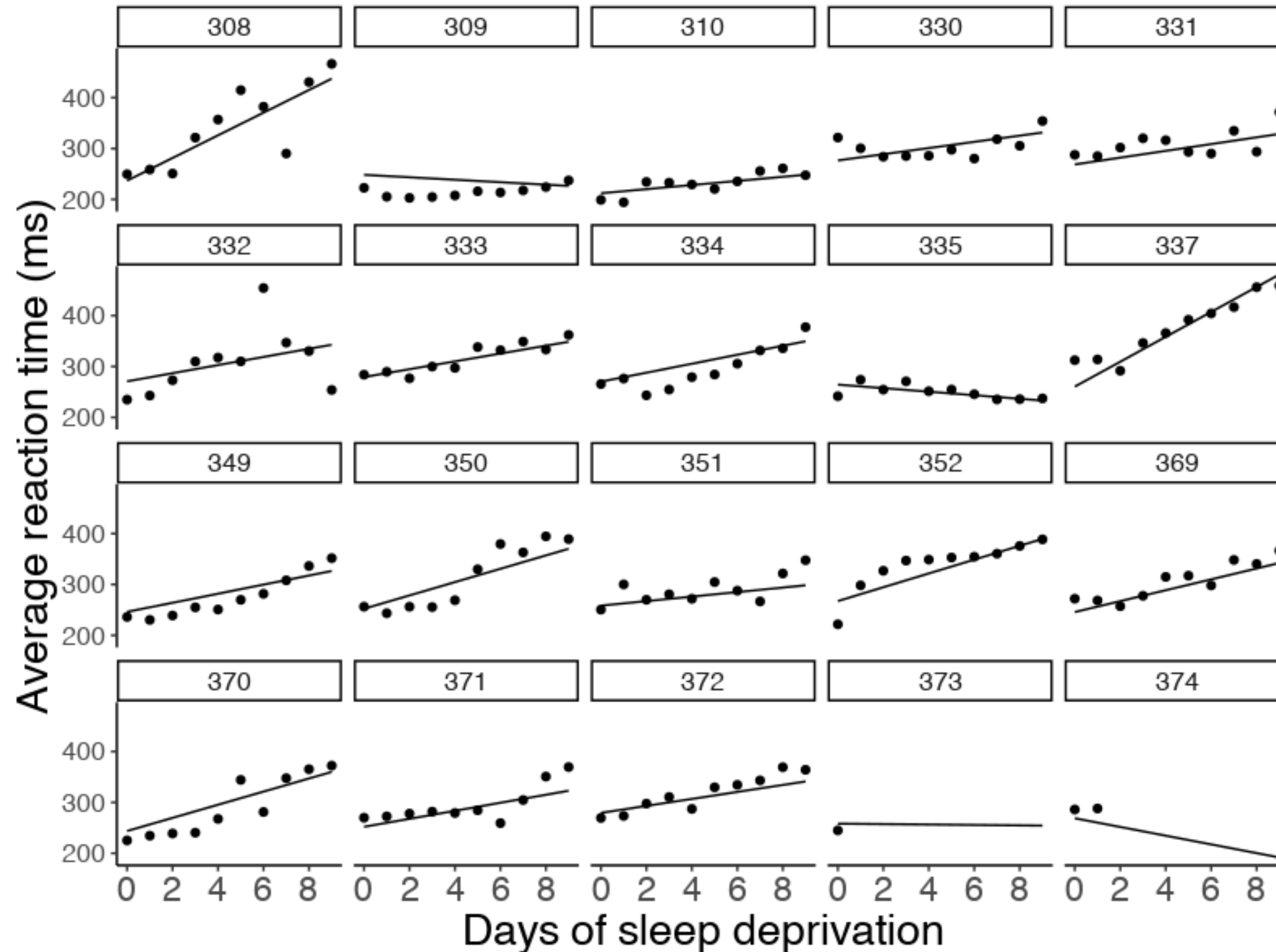
b) Visualize the model predictions



10 random samples from the posterior distribution

b) Visualize the model predictions

if you're feeling fancy

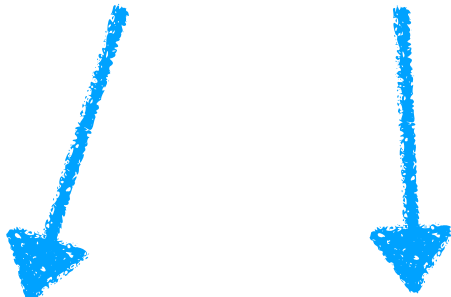


4. Interpret the model parameters

4. Interpret the model parameters

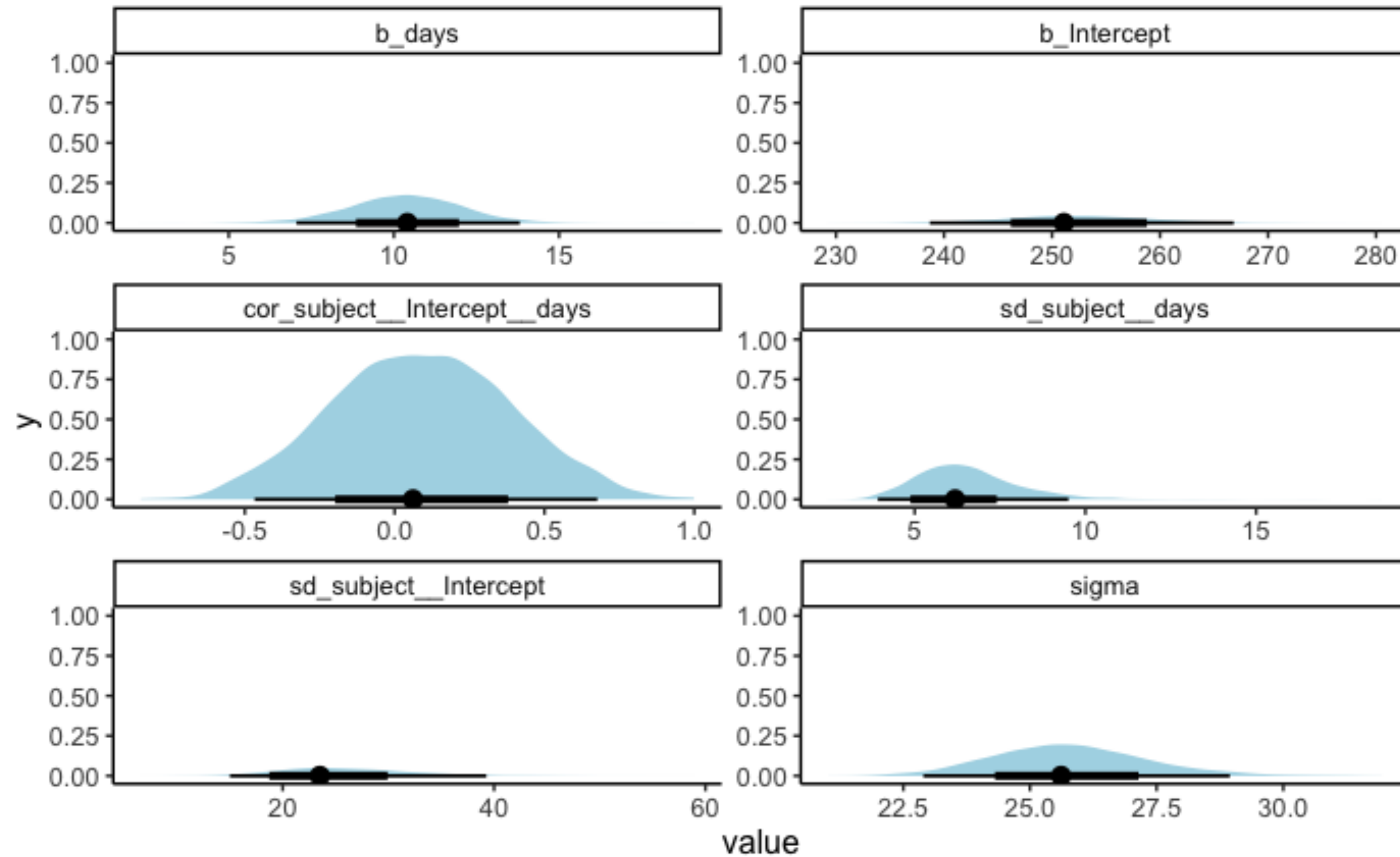
```
1 fit.brm_sleep %>%
2   tidy(conf.method = "HPDinterval")
```

95% highest
density interval



effect	component	group	term	estimate	std.error	conf.low	conf.high
fixed	cond	NA	(Intercept)	252.39	7.00	238.69	266.82
fixed	cond	NA	days	10.34	1.72	7.05	13.81
ran_pars	cond	subject	sd__(Intercept)	26.14	6.37	15.00	39.27
ran_pars	cond	subject	sd__days	6.55	1.54	3.92	9.50
ran_pars	cond	subject	cor__(Intercept).days	0.09	0.30	-0.47	0.68
ran_pars	cond	Residual	sd__Observation	25.80	1.54	22.90	28.95

4. Interpret the model parameters



Posterior distribution for most parameters

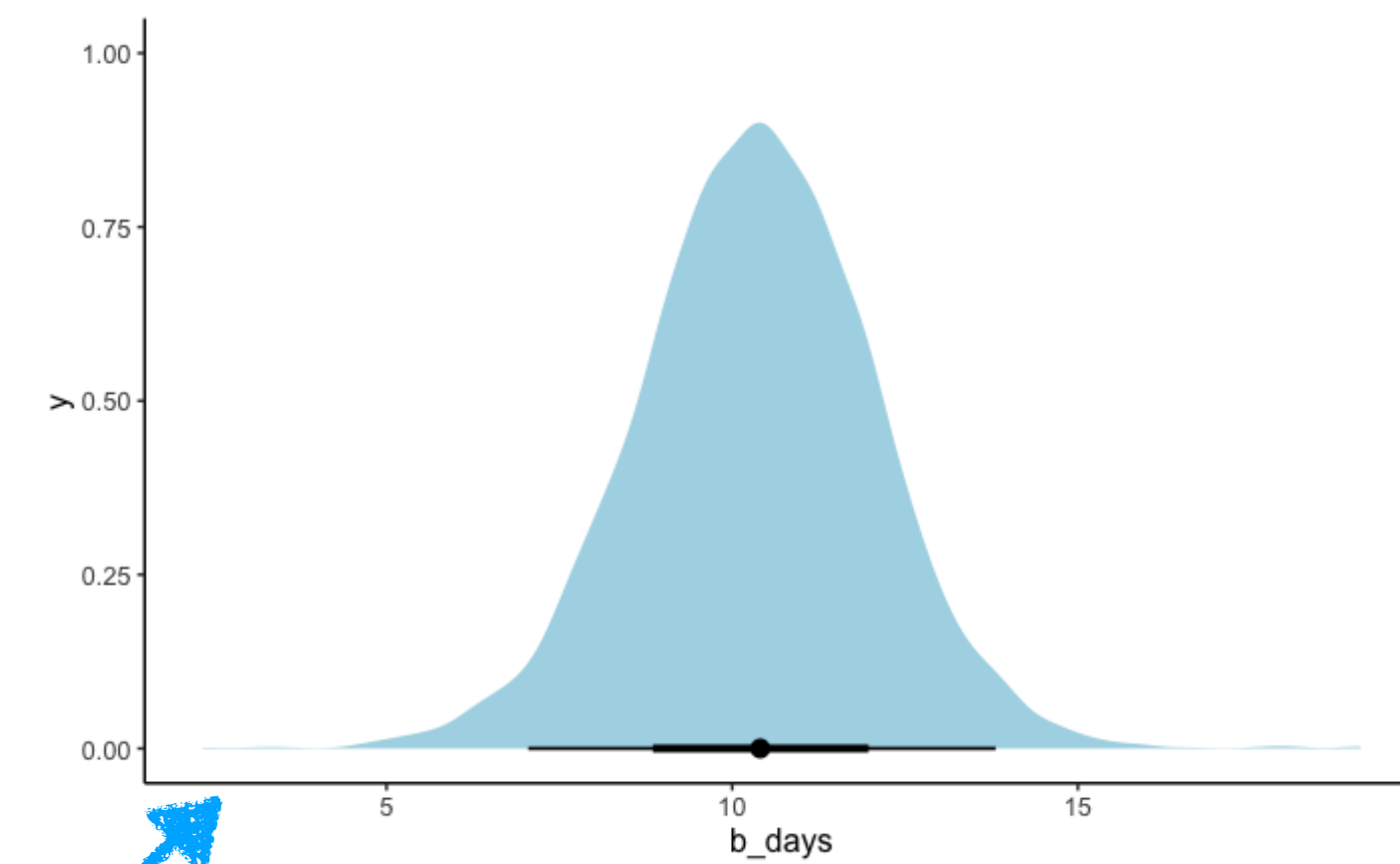
5. Test specific hypotheses

5. Test specific hypotheses

Did reaction times increase with the number of days of sleep deprivation?

```
1 fit.brm_sleep %>%  
2 summary()
```

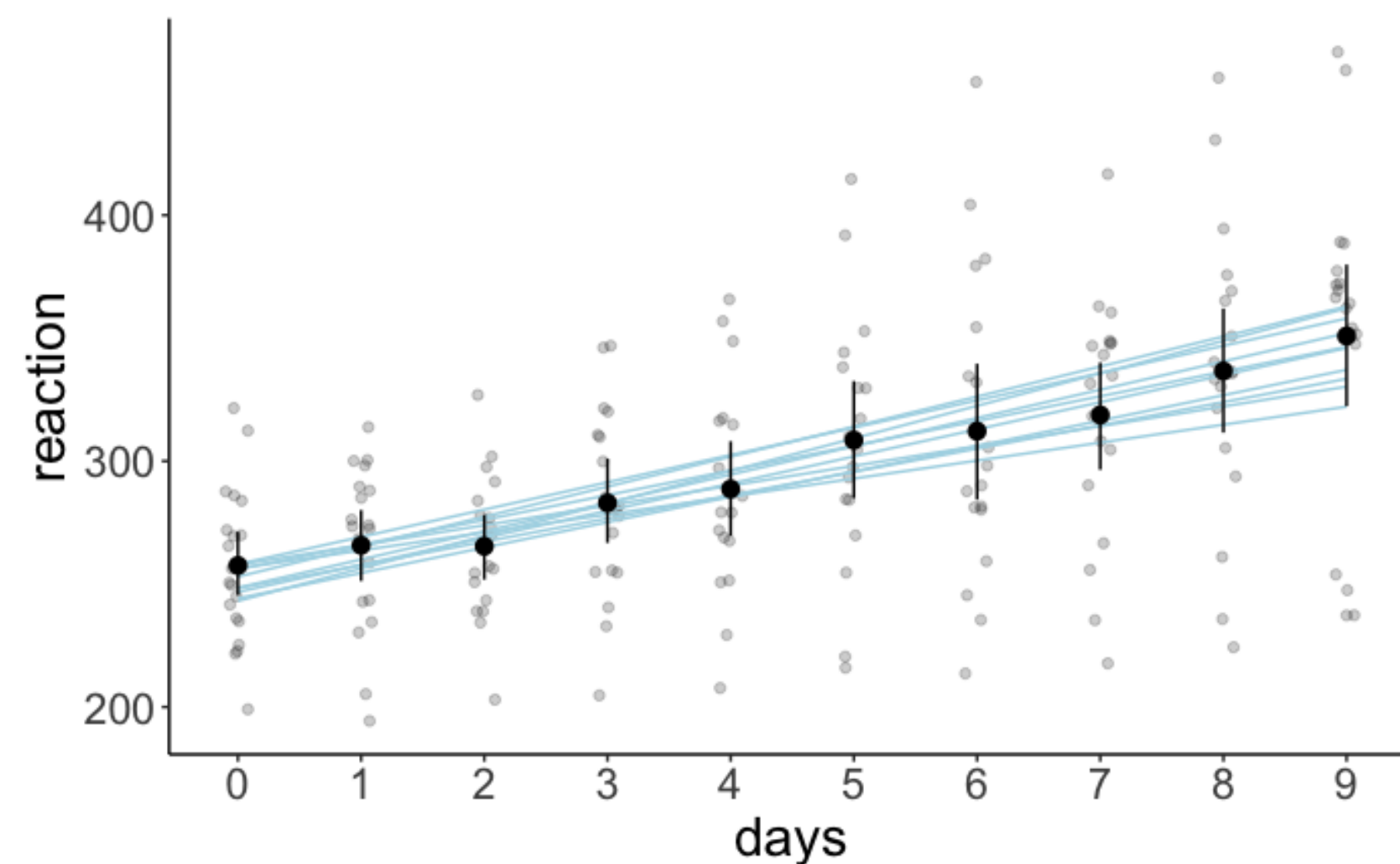
```
Family: gaussian  
Links: mu = identity; sigma = identity  
Formula: reaction ~ 1 + days + (1 + days | subject)  
Data: df.sleep (Number of observations: 183)  
Samples: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;  
total post-warmup samples = 4000  
  
Group-Level Effects:  
~subject (Number of levels: 20)  
              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS  
sd(Intercept)    26.18      6.25    15.65    40.54 1.00    1879    2463  
sd(days)          6.59      1.53     4.14    10.13 1.00    1145    1625  
cor(Intercept,days) 0.09      0.29    -0.46     0.67 1.00     993    1526  
  
Population-Level Effects:  
              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS  
Intercept    252.18      6.86    238.47    265.42 1.00    1826    2766  
days         10.46      1.69     7.13    13.78 1.00    1203    1782  
  
Family Specific Parameters:  
              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS  
sigma         25.77      1.57    22.93    29.14 1.00    3864    2773  
  
Samples were drawn using sampling(NUTS). For each parameter, Eff.Sample  
is a crude measure of effective sample size, and Rhat is the potential  
scale reduction factor on split chains (at convergence, Rhat = 1).
```



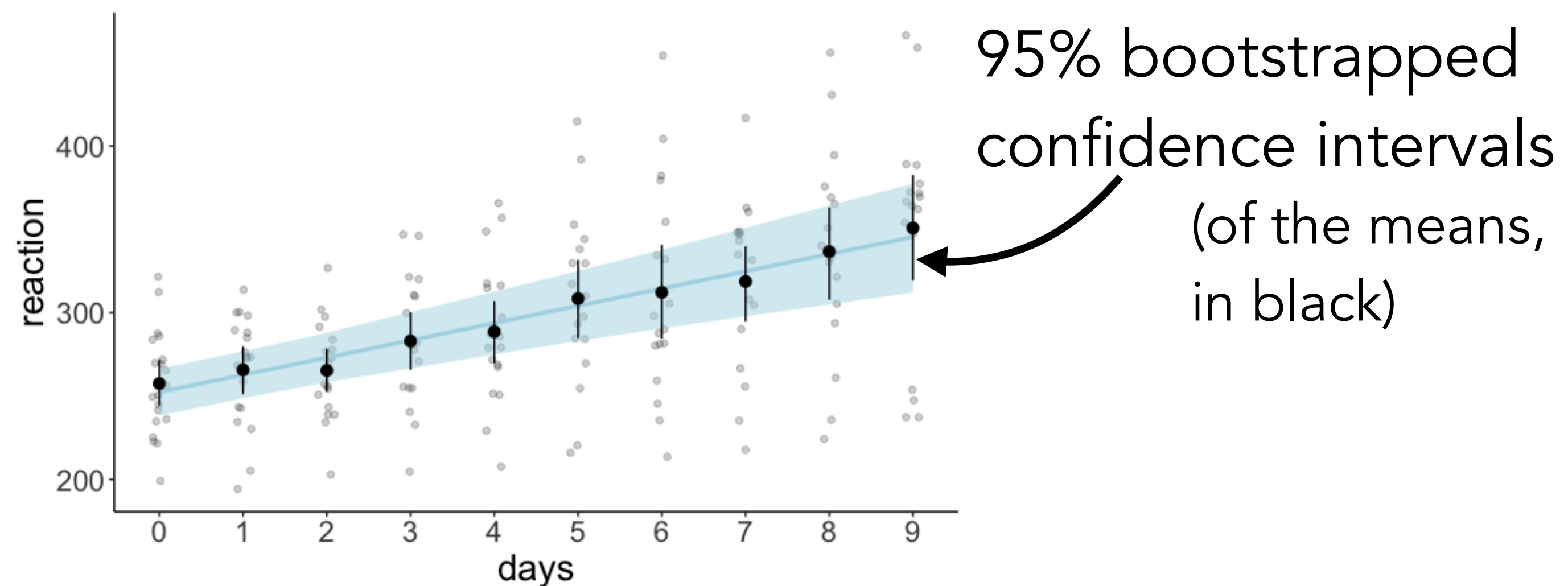
6. Report results

6. Report results

10 draws from the posterior



credible interval (in blue)



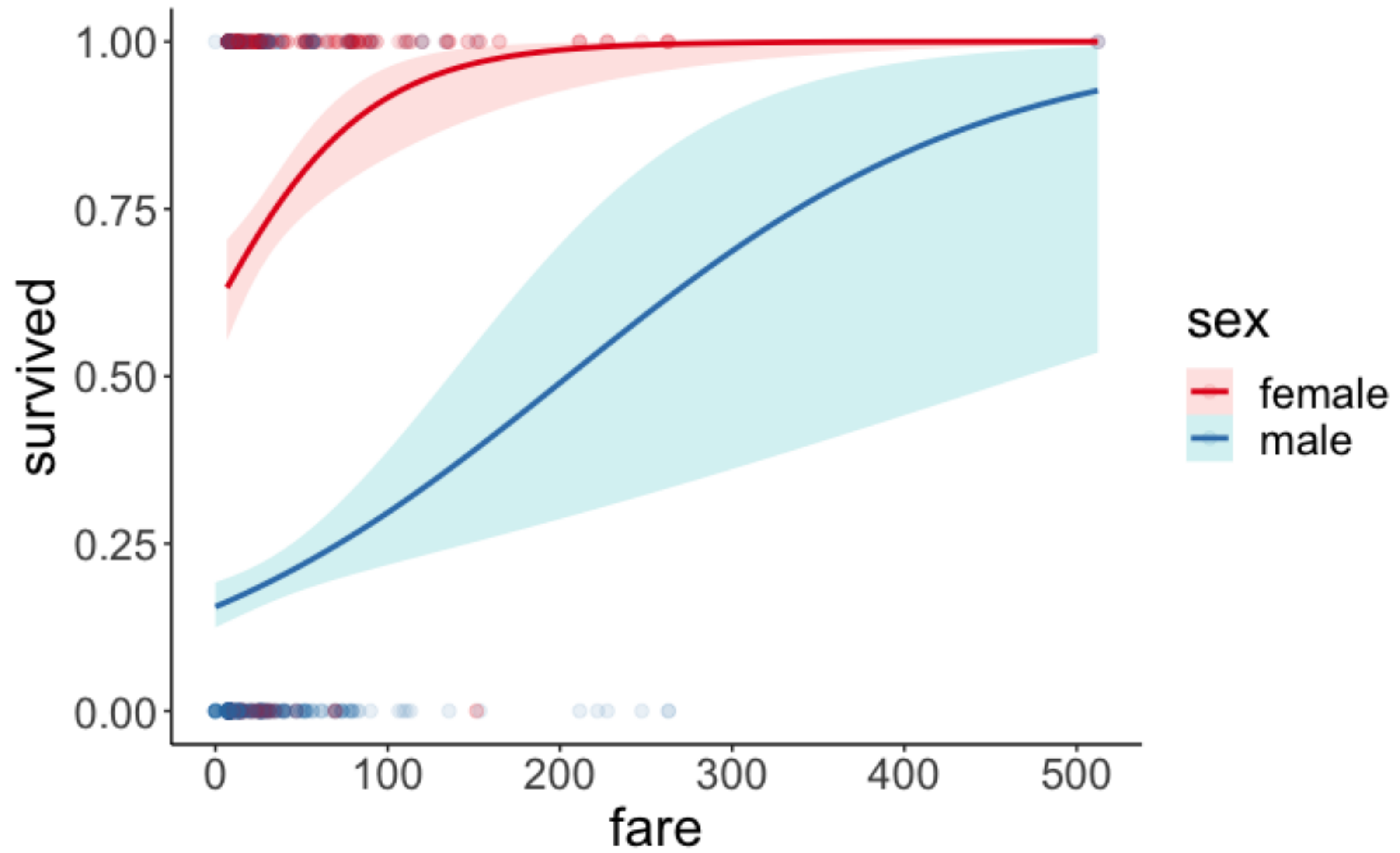
With each day of sleep deprivation, the reaction time increased by 10.5ms (95% HDI: 7.13, 13.78).

Recipe for Bayesian analysis with `brms`

1. Visualize the data
2. Specify and fit the model
3. Model evaluation
 - a) Did the inference work?
 - b) Visualize model predictions
4. Interpret the model parameters
5. Test specific hypotheses
6. Report results

Titanic data

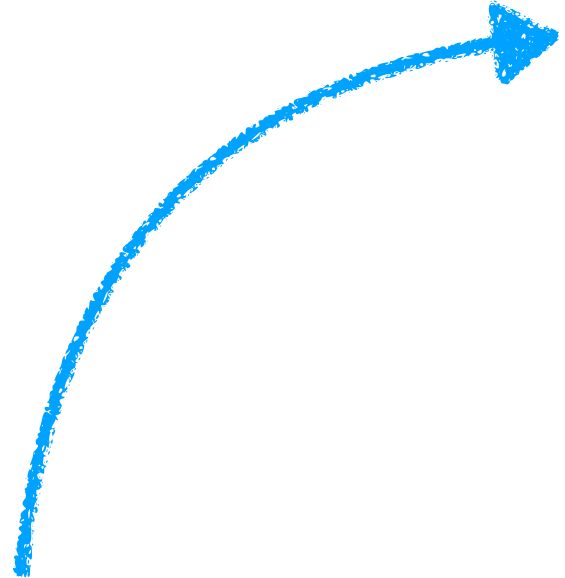
1. Visualize the data



2. Specify and fit the model

1. Specify and fit the model

```
1 fit.brm_titanic = brm(formula = survived ~ 1 + fare * sex,  
2 family = "bernoulli",  
3 data = df.titanic,  
4 file = "cache/brm_titanic",  
5 seed = 1)
```



**just need to
change the family**

3. Model evaluation

a) Did the inference work?

```
1 fit.brm_titanic %>%  
2   summary()
```

```
Family: bernoulli  
Links: mu = logit  
Formula: survived ~ 1 + fare * sex  
Data: df.titanic (Number of observations: 891)  
Samples: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;  
         total post-warmup samples = 4000
```

Population-Level Effects:

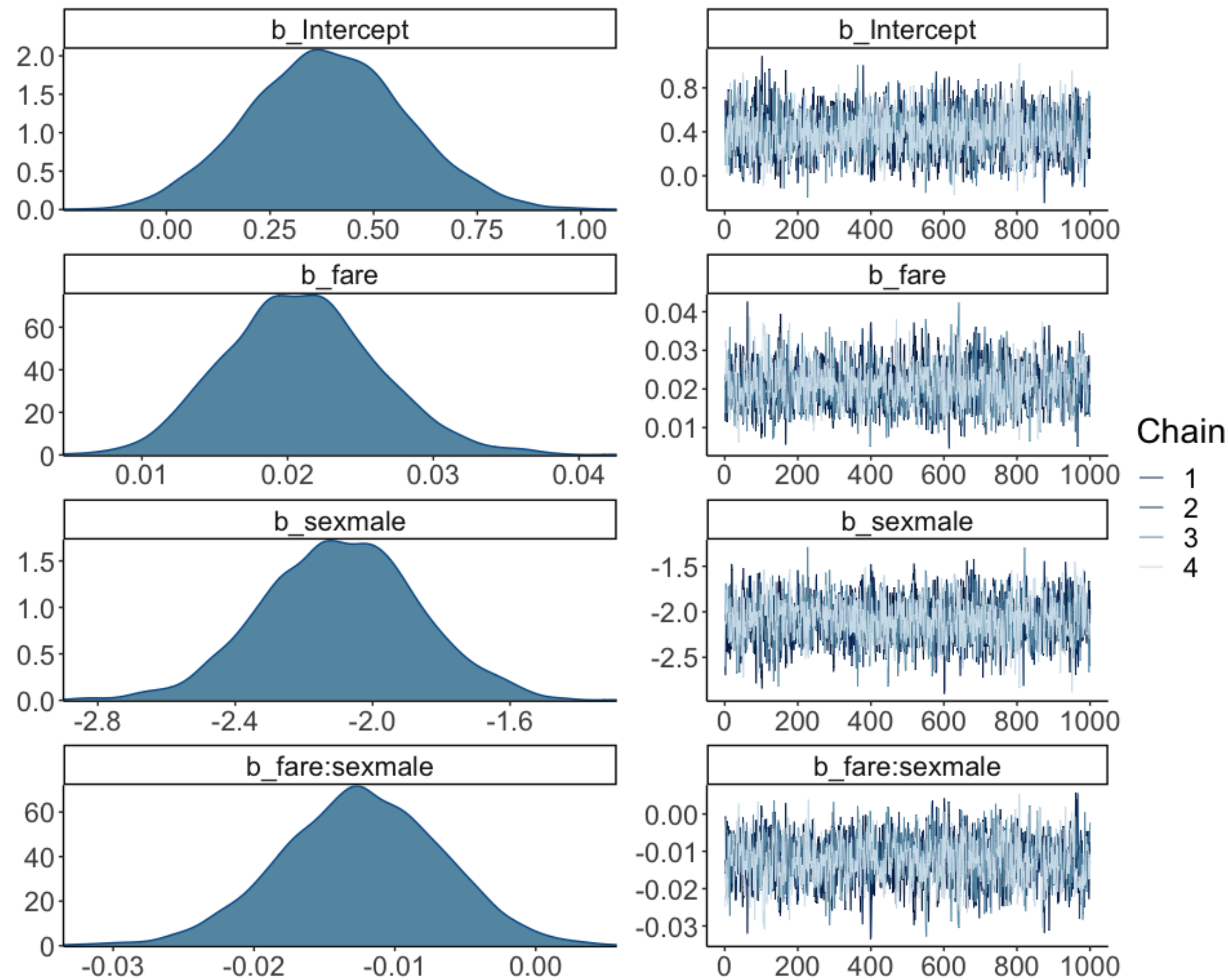
	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	0.39	0.19	0.03	0.76	1.00	2010	2625
fare	0.02	0.01	0.01	0.03	1.00	1545	2124
sexmale	-2.09	0.23	-2.54	-1.65	1.00	1754	1984
fare:sexmale	-0.01	0.01	-0.02	-0.00	1.00	1479	2041

Samples were drawn using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

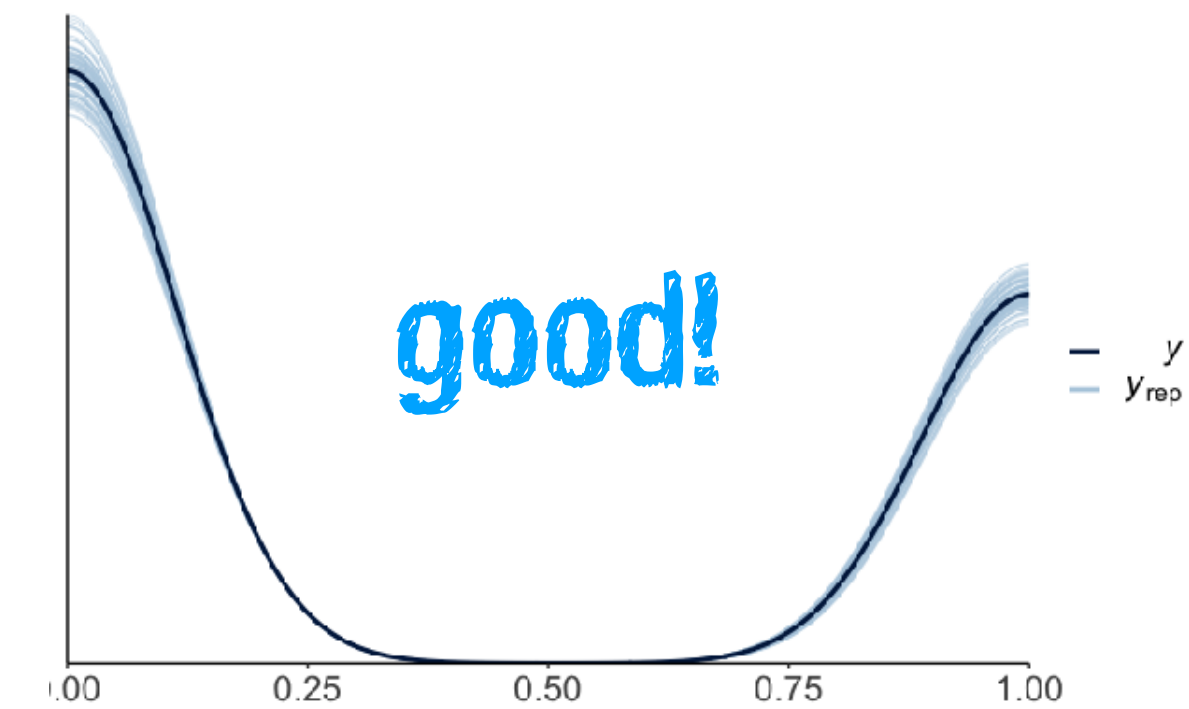
looks good

a) Did the inference work?

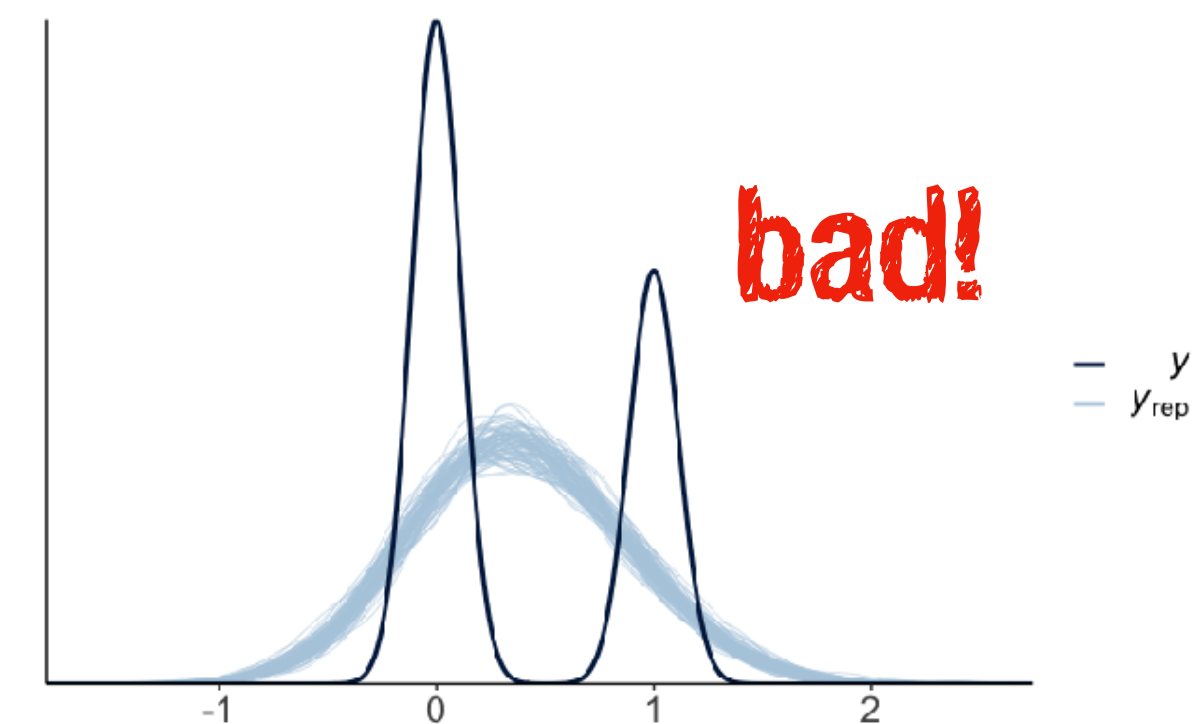
```
1 fit.brm_titanic %>%  
2   plot()
```



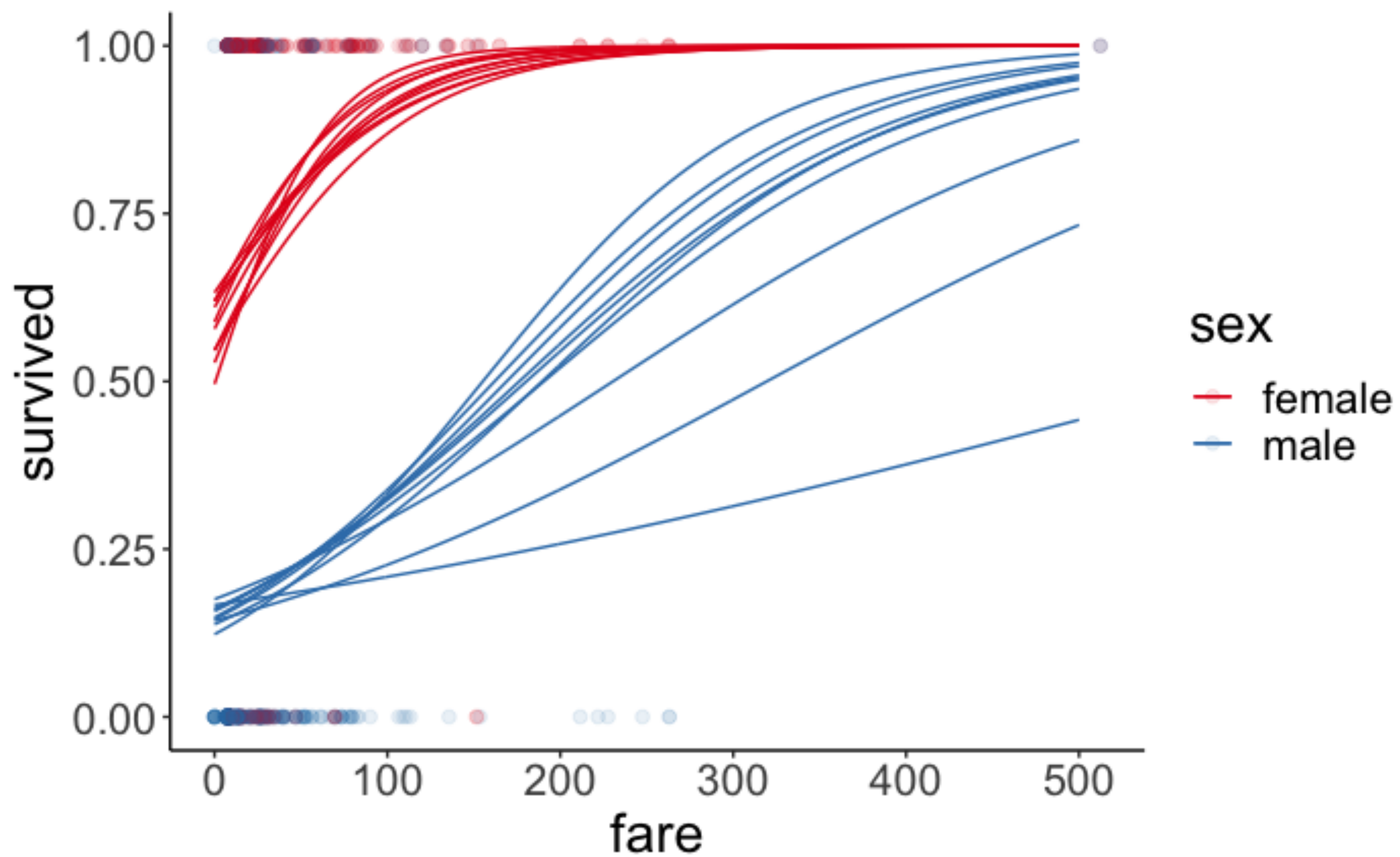
```
1 pp_check(fit.brm_titanic,  
2          nsamples = 100)
```



model with Gaussian family



b) Visualize the model predictions



4. Interpret the model parameters

4. Interpret the model parameters

```
Family: bernoulli
Links: mu = logit
Formula: survived ~ 1 + fare * sex
Data: df.titanic (Number of observations: 891)
Samples: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup samples = 4000
```

Population-Level Effects:

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	0.39	0.19	0.03	0.76	1.00	2010	2625
fare	0.02	0.01	0.01	0.03	1.00	1545	2124
sexmale	-2.09	0.23	-2.54	-1.65	1.00	1754	1984
fare:sexmale	-0.01	0.01	-0.02	-0.00	1.00	1479	2041

Samples were drawn using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

log odds

4. Interpret the model parameters



```
1 fit.brm_titanic %>%  
2   ggpredict(terms = c("fare [0:500]", "sex"))
```

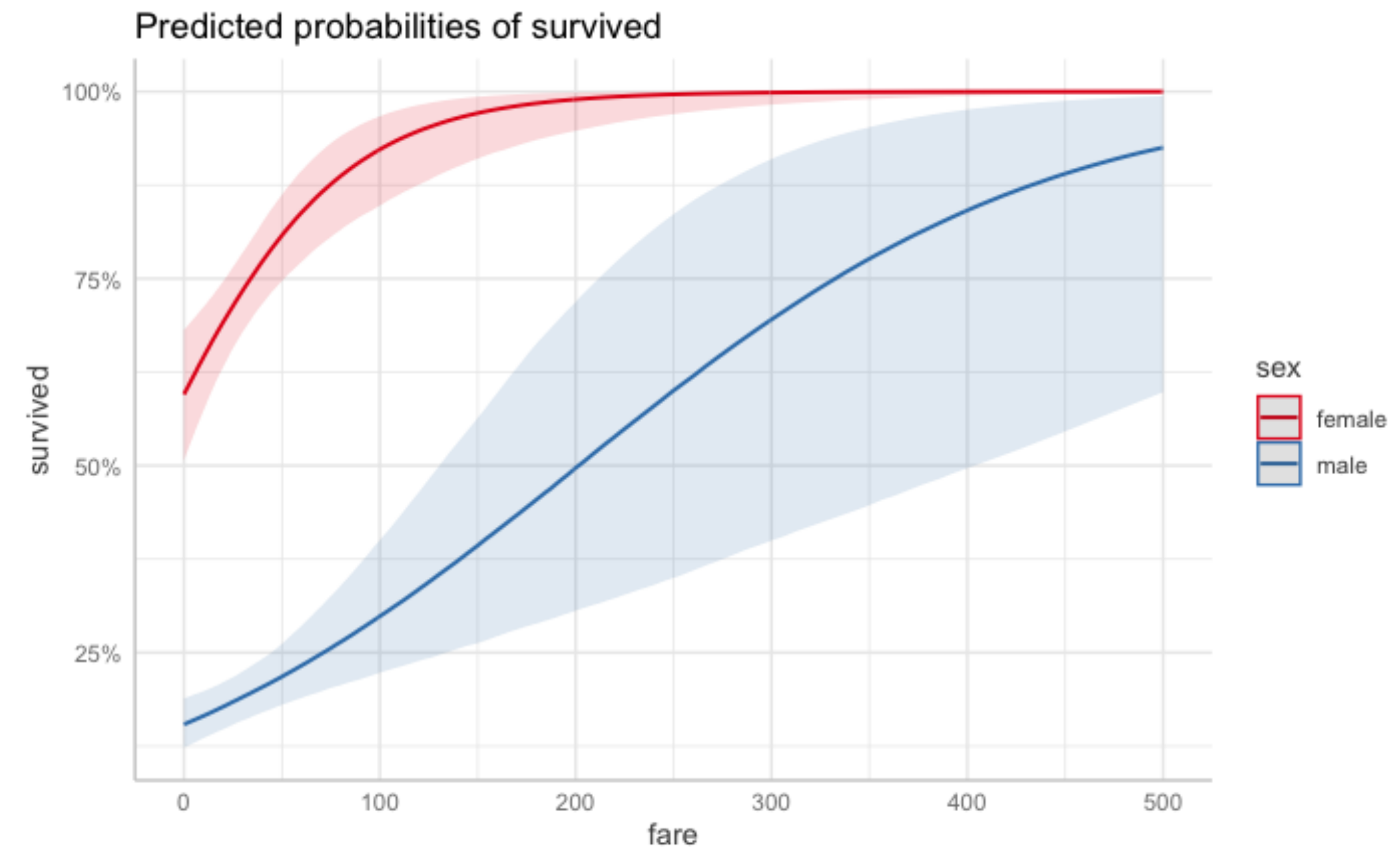
```
# Predicted probabilities of survived  
# x = fare  
  
# sex = female  


| x   | Predicted | 95% CI       |
|-----|-----------|--------------|
| 0   | 0.60      | [0.51, 0.68] |
| 83  | 0.89      | [0.82, 0.95] |
| 167 | 0.98      | [0.93, 1.00] |
| 250 | 1.00      | [0.97, 1.00] |
| 333 | 1.00      | [0.99, 1.00] |
| 500 | 1.00      | [1.00, 1.00] |

  
# sex = male  


| x   | Predicted | 95% CI       |
|-----|-----------|--------------|
| 0   | 0.15      | [0.12, 0.19] |
| 83  | 0.27      | [0.21, 0.35] |
| 167 | 0.43      | [0.28, 0.62] |
| 250 | 0.60      | [0.35, 0.84] |
| 333 | 0.75      | [0.43, 0.94] |
| 500 | 0.93      | [0.60, 0.99] |


```



5. Test specific hypotheses

5. Test specific hypotheses

Were women more likely to survive than men?

```
1 fit.brm_titanic %>%
2   emmeans(specs = pairwise ~ sex,
3           type = "response")
```

The odds of female passengers surviving was 12 times higher than the odds of male passengers surviving.

$$\frac{\left(\frac{p_f}{1-p_f}\right)}{\left(\frac{p_m}{1-p_m}\right)}$$

NOTE: Results may be misleading due to involvement in interactions

\$emmeans

sex	response	lower.HPD	upper.HPD
female	0.743	0.69	0.795
male	0.194	0.16	0.225

Point estimate displayed: median

Results are back-transformed from the logit scale

HPD interval probability: 0.95

\$contrasts

contrast	odds.ratio	lower.HPD	upper.HPD
female / male	12.1	8.39	16.6

Point estimate displayed: median

Results are back-transformed from the log odds ratio scale

HPD interval probability: 0.95

5. Test specific hypotheses

Was the effect of fare on survival different for men vs women?

```
1 fit.brm_titanic %>%  
2   emtrends(specs = pairwise ~ sex,  
3           var = "fare")
```

```
$emtrends  
sex    fare.trend lower.HPD upper.HPD  
female 0.02083    0.01129    0.0316  
male   0.00845    0.00385    0.0135
```

Point estimate displayed: median
HPD interval probability: 0.95

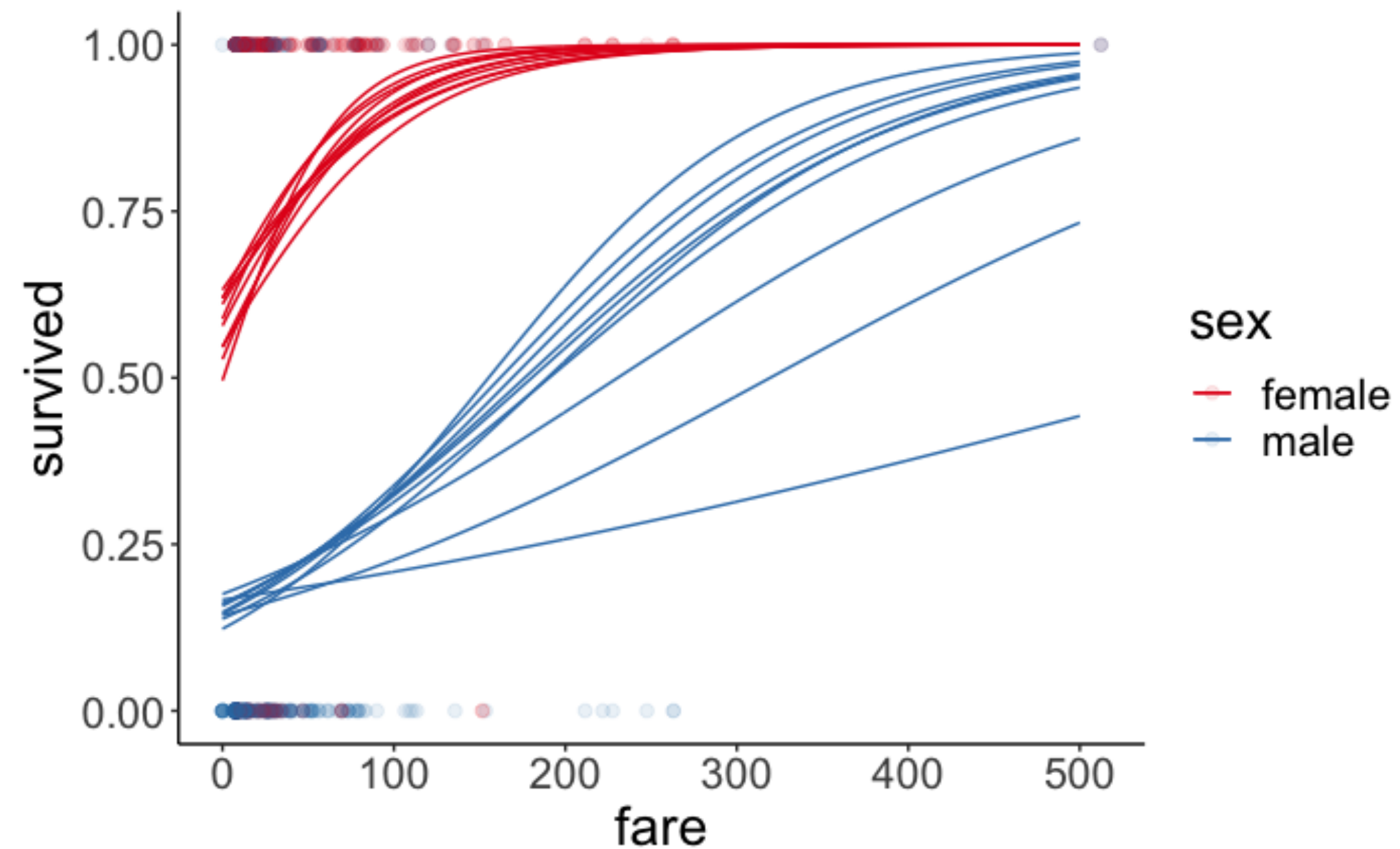
```
$contrasts  
contrast      estimate lower.HPD upper.HPD  
female - male 0.0124    0.000884    0.0232
```

Point estimate displayed: median
HPD interval probability: 0.95

the chance of survival
increased more with fare
for female than male
passengers

6. Report results

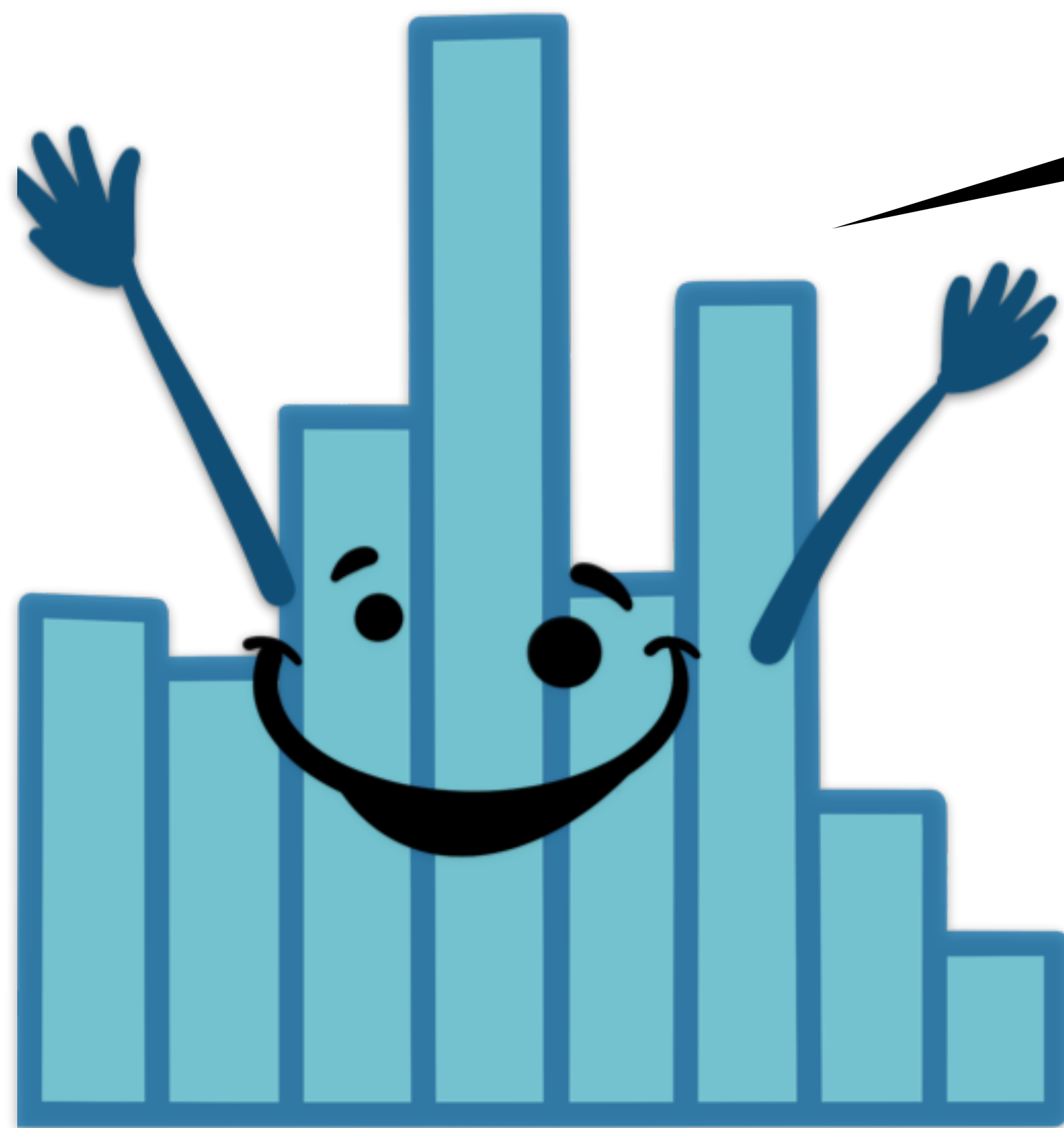
6. Report results



Female passengers were more likely to survive (74.3%) than male passengers (19.4%). The estimated odds of women surviving was 12.1 95% CrI [8.4, 16.6] times higher than men.

The chance of survival increased more with fare for female compared to male passengers. The difference in slopes on the log odds scale was 0.01 [0, 0.02].

02:00
stretch break!



Going beyond

Evidence for the null hypothesis

Evidence for the null hypothesis


[Front Psychol.](#) 2014; 5: 781. PMCID: PMC4114196
Published online 2014 Jul 29. doi: [10.3389/fpsyg.2014.00781](https://doi.org/10.3389/fpsyg.2014.00781) PMID: [25120503](https://pubmed.ncbi.nlm.nih.gov/25120503/)

Using Bayes to get the most out of non-significant results

[Zoltan Dienes](#)*

► [Author information](#) ► [Article notes](#) ► [Copyright and License information](#) [Disclaimer](#)

[HTML] [Using Bayes to get the most out of non-significant results](#)

[Z Dienes](#) - [Frontiers in psychology, 2014 - frontiersin.org](#)

No scientific conclusion follows automatically from a statistically non-significant result, yet people routinely use non-significant results to guide conclusions about the status of theories (or the effectiveness of practices). To know whether a non-significant result counts against a theory, or if it just indicates data insensitivity, researchers must use one of: power, intervals (such as confidence or credibility intervals), or else an indicator of the relative evidence for one theory over another, such as a Bayes factor. I argue Bayes factors allow theory to be ...

☆ ⓘ [Cited by 966](#) [Related articles](#) [All 14 versions](#) [Web of Science: 583](#) [Import into BibTeX](#) ⓘ

2468 now

- There is nothing special about H_0 compared to H_1 in Bayesian inference
- We can get evidence of H_0 over H_1 (e.g. using the Bayes factor approach)

Bayes factor

$$BF_{01} = \frac{p(D | H_0)}{p(D | H_1)}$$

probability of the data
given H_0

probability of the data
given H_1

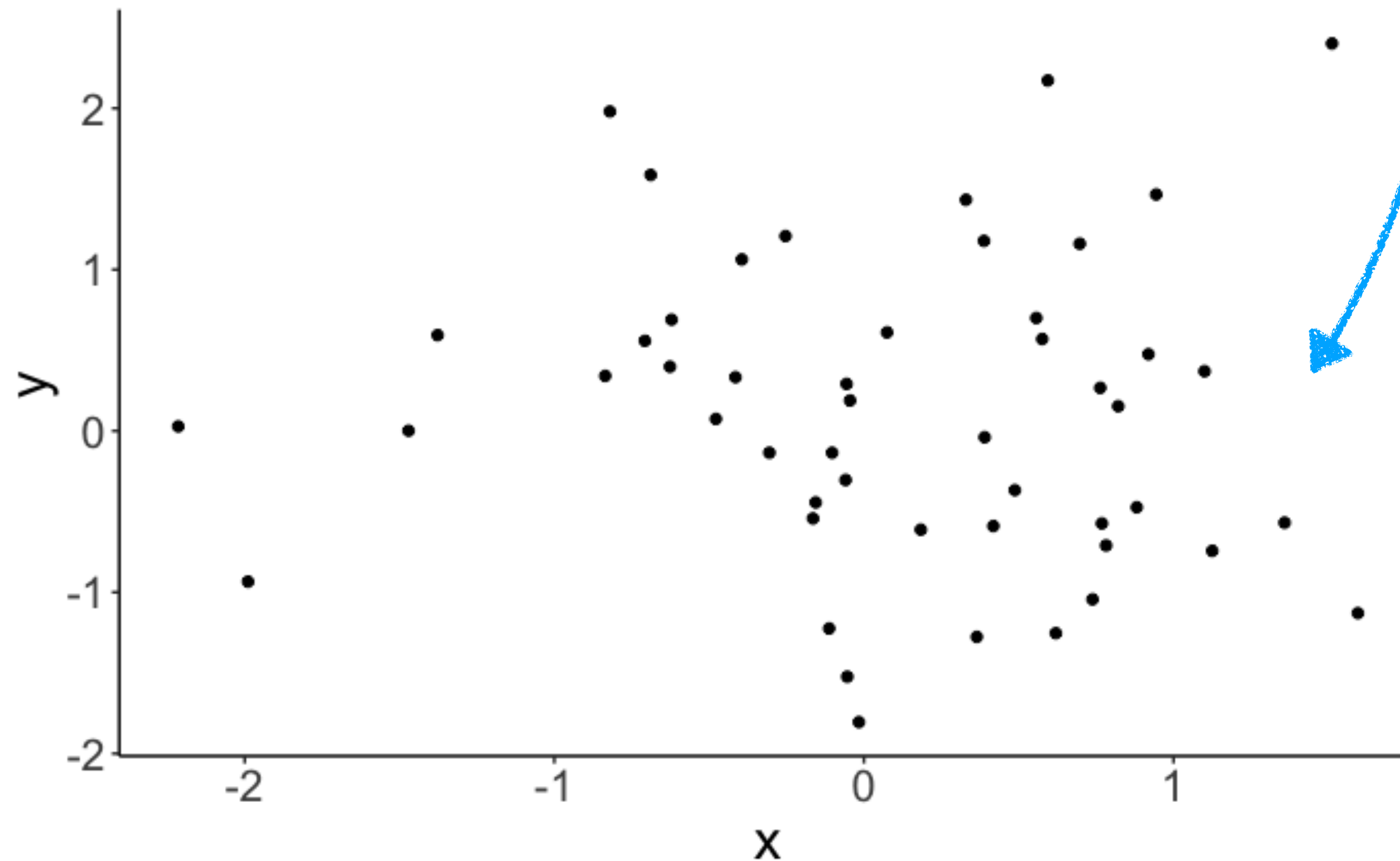
check this out

<https://vuorre.netlify.com/post/2017/03/21/bayes-factors-with-brms/>

Approximate LOO

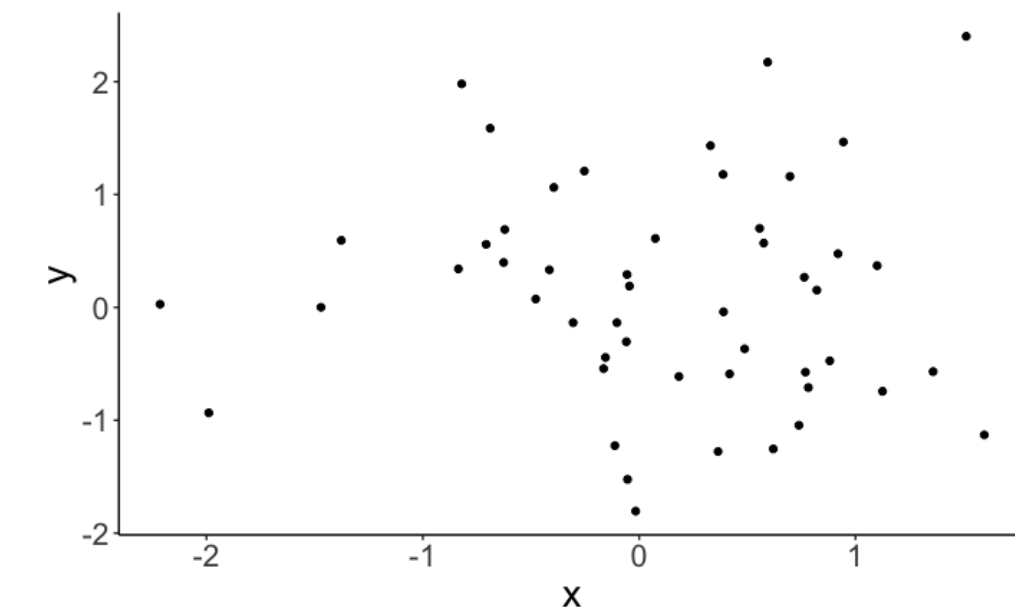
Evidence for the null hypothesis

```
1 set.seed(1)
2 df.loo = tibble(x = rnorm(n = 50),
3                  y = rnorm(n = 50))
4
5 ggplot(data = df.loo,
6         mapping = aes(x = x,
7                       y = y)) +
8   geom_point()
```



no relationship
between x and y

Evidence for the null hypothesis

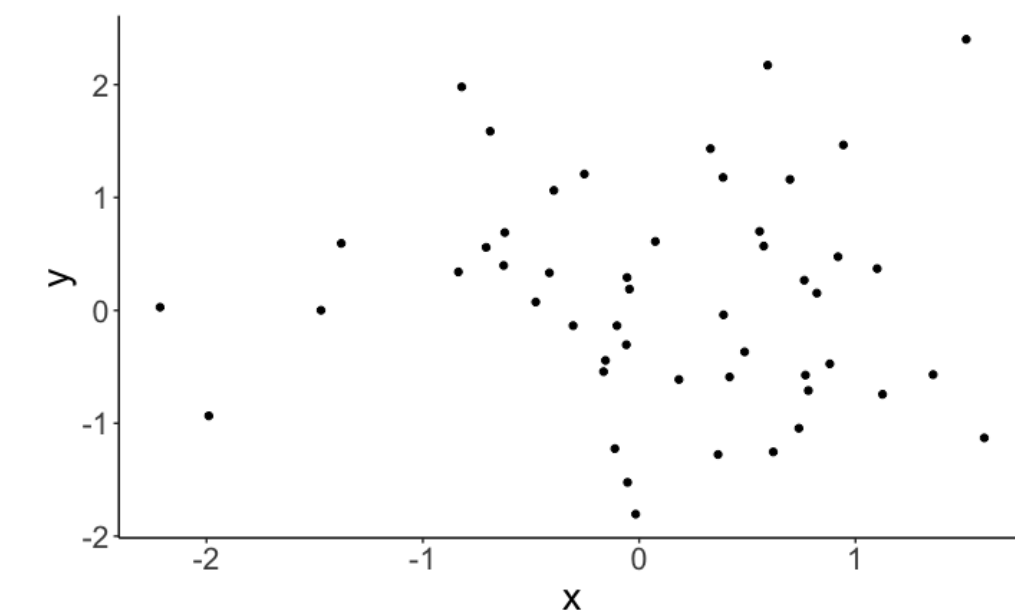


```
1 fit.lm_loo = lm(formula = y ~ 1 + x,  
2                 data = df.loo)  
3  
4 fit.lm_loo %>%  
5   summary()
```

```
Call:  
lm(formula = y ~ 1 + x, data = df.loo)  
  
Residuals:  
      Min       1Q   Median       3Q      Max   
-4.2185 -0.6735  0.0018  0.6734  4.2428   
  
Coefficients:  
              Estimate Std. Error t value Pr(>|t|)      
(Intercept)  0.0006437   0.0031639   0.203   0.839      
x            -0.0019184   0.0031541  -0.608   0.543      
  
Residual standard error: 1.001 on 99998 degrees of freedom  
Multiple R-squared:  3.7e-06, Adjusted R-squared:  -6.301e-06  
F-statistic:  0.37 on 1 and 99998 DF,  p-value: 0.543
```

cannot reject the H_0 that the reduction in error due to x is what one would have expected by chance

Evidence for the null hypothesis



```
1 fit.brm_loo1 = brm(formula = y ~ 1, data = df.loo)
2
3 fit.brm_loo2 = brm(formula = y ~ 1 + x, data = df.loo)
4
5 fit.brm_loo1 = add_criterion(fit.brm_loo1, criterion = "loo")
6
7 fit.brm_loo2 = add_criterion(fit.brm_loo2, criterion = "loo")
```

approximate
leave-one-out
cross-validation

```
loo_compare(fit.brm_loo1, fit.brm_loo2)
```

	elpd_diff	se_diff
fit.brm_loo1	0.0	0.0
fit.brm_loo2	-1.1	0.5

Difference in Expected Log Predictive Density

In plain English: It tells you how much worse a model is at predicting new data compared to the best model in your list.

```
model_weights(fit.brm_loo1, fit.brm_loo2)
```

fit.brm_loo1	fit.brm_loo2
99.99999	0.00001

The Rule of Thumb: If the absolute value of elpd_diff is more than 2 times larger than the se_diff, you can be confident that the winning model is genuinely better.

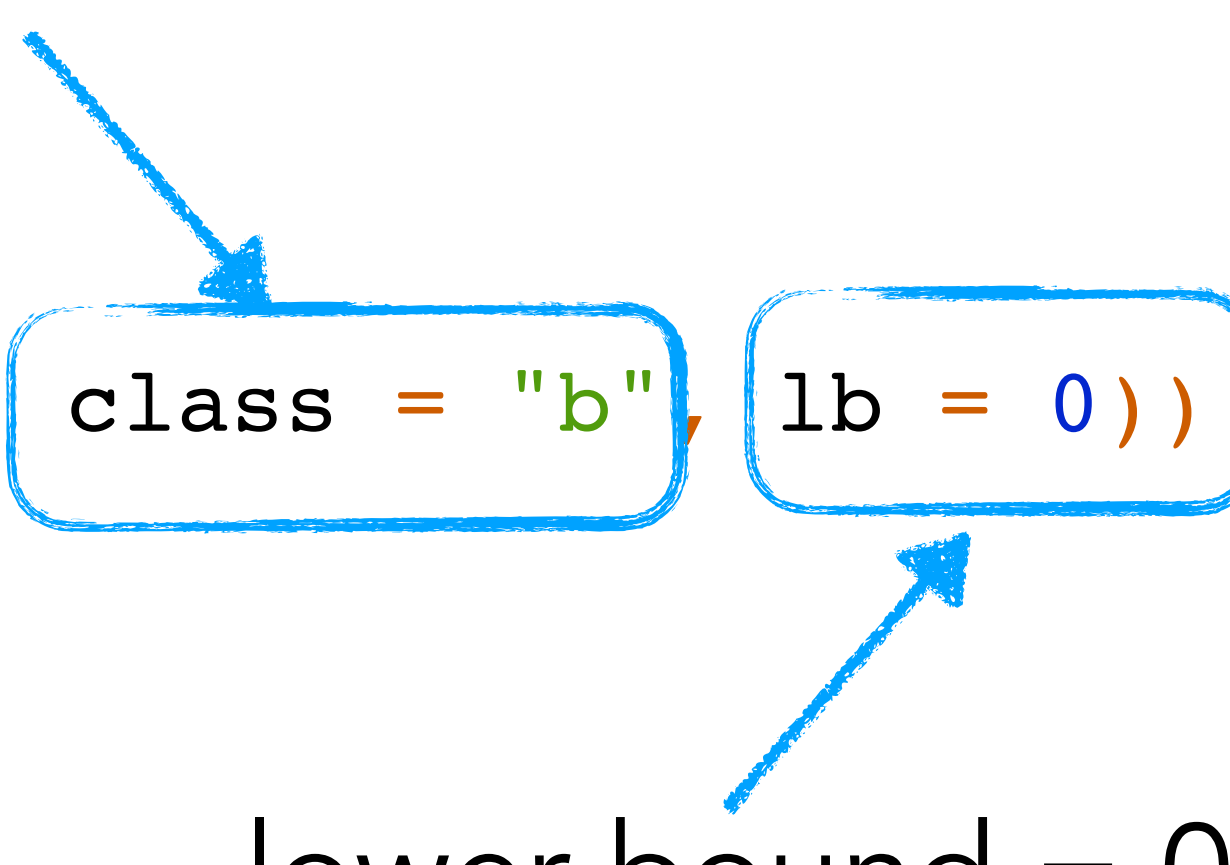
I want only positive coefficients!

I only want positive coefficients!

```
1 brm(formula = how_much_i_love_stats ~ 1 + nilam + tobi + justin + cynthia +  
      verity + shashanka + alvin,  
2     data = df.stats_love)
```

coefficients in the model

```
1 # priors  
2 priors = c(set_prior("normal(0,10)",  
                      class = "b",  
                      lb = 0))
```



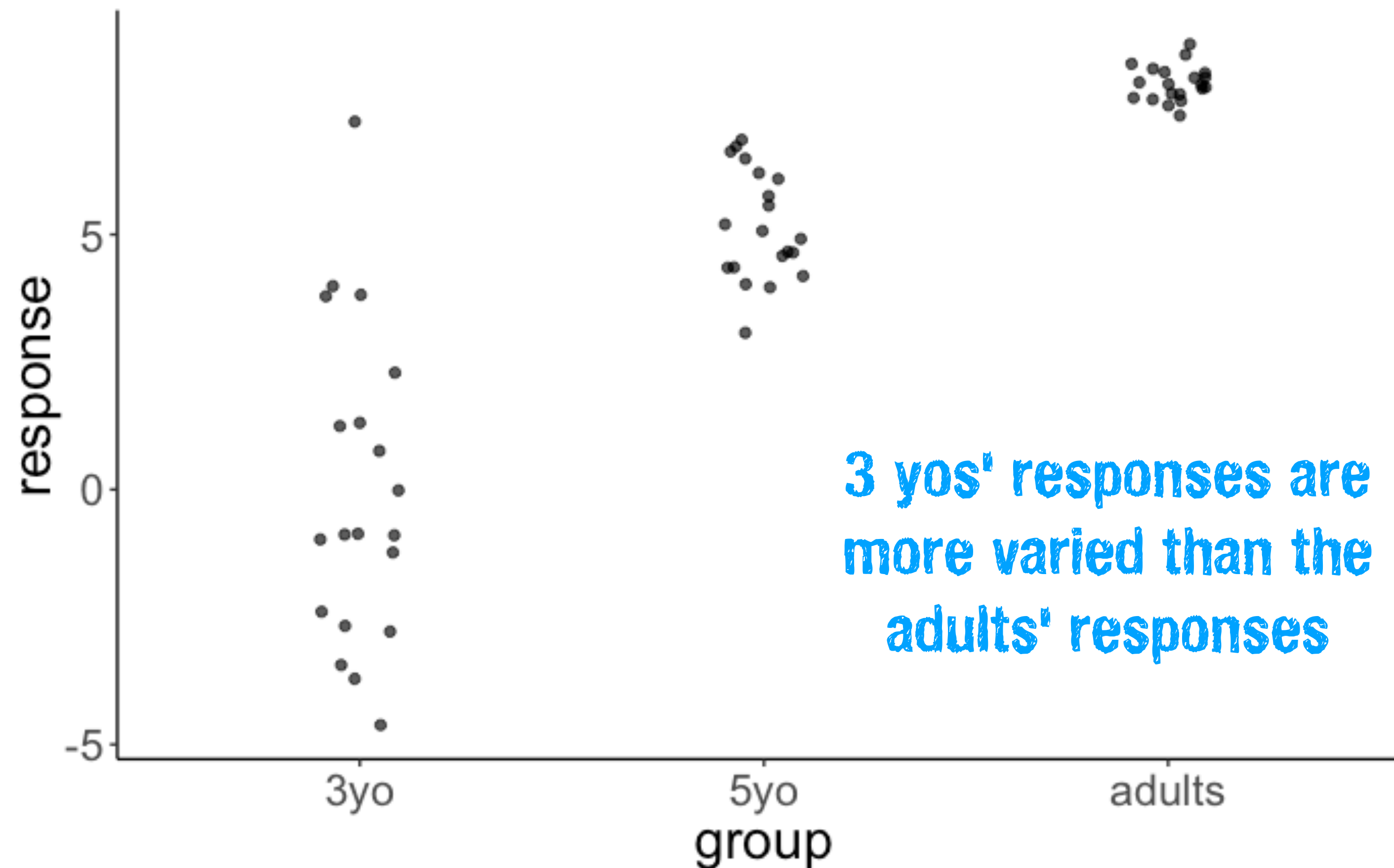
lower bound = 0

```
1 brm(formula = how_much_i_love_stats ~ 1 + nilam + tobi + justin + cynthia +  
      verity + shashanka + alvin,  
2     prior = priors,  
3     data = df.stats_love)
```

Dealing with unequal variance

Unequal variance aka heteroscedasticity

```
1 df.variance = tibble(group = rep(c("3yo", "5yo", "adults"), each = 20),  
2                       response = rnorm(n = 60,  
3                                       mean = rep(c(0, 5, 8), each = 20),  
4                                       sd = rep(c(3, 1.5, 0.3), each = 20)))
```



Unequal variance aka heteroscedasticity

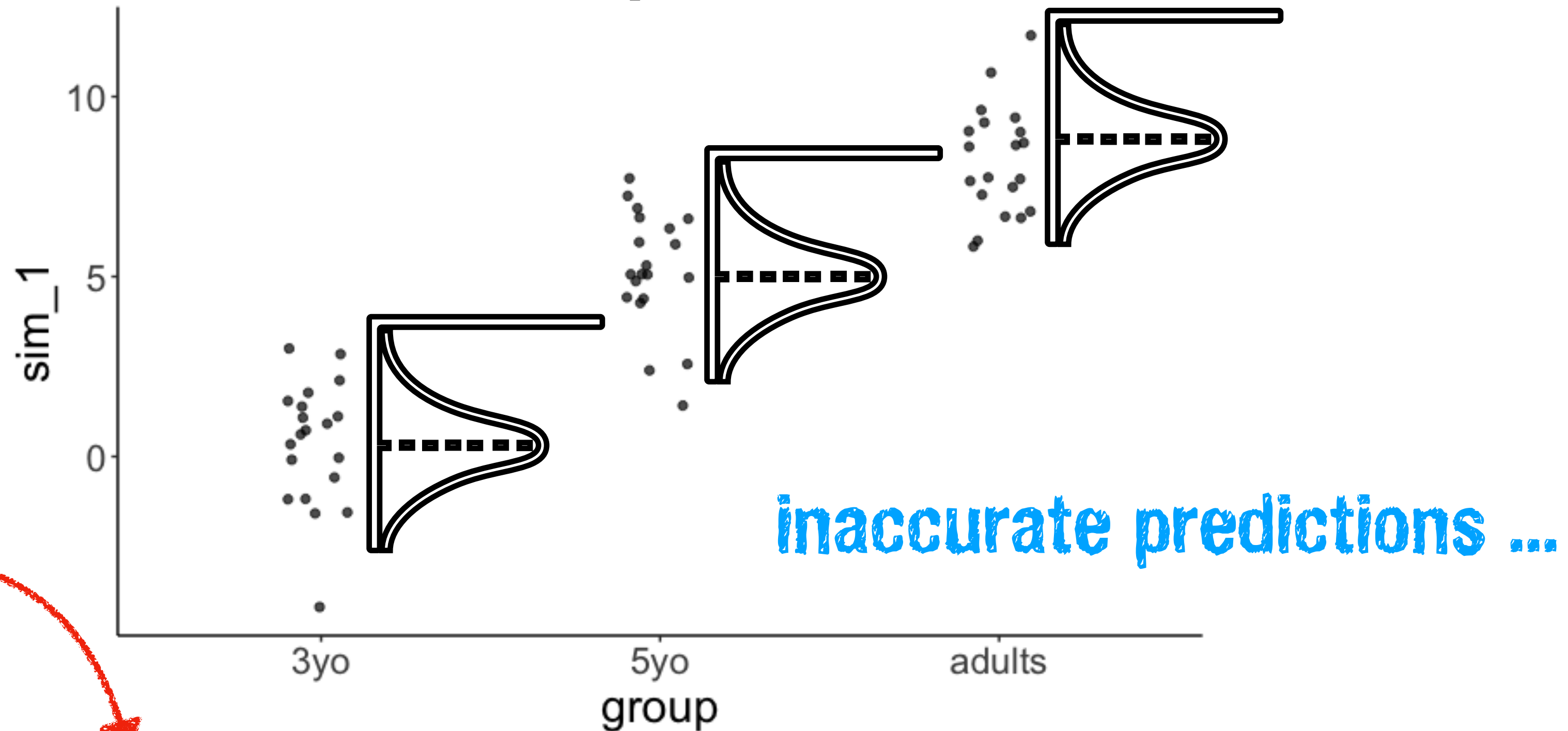
```
1 fit.lm1 = lm(formula = response ~ 1 + group,  
2             data = df.variance)  
3  
4 fit.lm1 %>%  
5   summary()
```

```
Call:  
lm(formula = response ~ 1 + group, data = df.variance)  
  
Residuals:  
      Min       1Q   Median       3Q      Max   
-4.6145 -0.8288 -0.0879  0.6315  7.2193  
  
Coefficients:  
              Estimate Std. Error t value Pr(>|t|)      
(Intercept) -0.005336   0.421618  -0.013    0.99      
group5yo     5.172810   0.596258   8.675 5.25e-12 ***  
groupadults  7.970655   0.596258  13.368 < 2e-16 ***  
---  
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
  
Residual standard error: 1.886 on 57 degrees of freedom  
Multiple R-squared:  0.7635,    Adjusted R-squared:  0.7552  
F-statistic: 91.99 on 2 and 57 DF,  p-value: < 2.2e-16
```

Unequal variance aka heteroscedasticity

```
1 fit.lm1 %>%
2   simulate() %>%
3   bind_cols(df.variance) %>%
4   ggplot(aes(x = group, y = sim_1)) +
5   geom_jitter(height = 0,
6               width = 0.1,
7               alpha = 0.7)
```

only one
standard
deviation
fitted



r.squared	adj.r.squared	sigma	statistic	p.value	df	logLik	AIC	BIC	deviance	df.residual
0.76	0.76	1.89	91.99	0	3	-121.65	251.3	259.68	202.65	57

Unequal variance aka heteroscedasticity

```
1 fit.brm1 = brm(formula = bf(response ~ group,  
2                               sigma ~ group),  
3               data = df.variance,  
4               file = "cache/brm1",  
5               seed = 1)
```

← modeling both the
means and variances

```
Family: gaussian  
Links: mu = identity; sigma = log  
Formula: response ~ group  
         sigma ~ group  
Data: df.variance (Number of observations: 60)  
Samples: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;  
         total post-warmup samples = 4000
```

Population-Level Effects:

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	-0.01	0.73	-1.41	1.51	1.01	1107	1072
sigma_Intercept	1.15	0.17	0.85	1.51	1.00	1991	1922
group5yo	5.18	0.77	3.60	6.65	1.00	1252	1327
groupadults	7.98	0.74	6.47	9.37	1.01	1110	1079
sigma_group5yo	-1.05	0.24	-1.51	-0.57	1.00	2249	2420
sigma_groupadults	-2.19	0.24	-2.66	-1.74	1.00	2171	2427

Samples were drawn using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Unequal variance aka heteroscedasticity

```
Family: gaussian
Links: mu = identity; sigma = log
Formula: response ~ group
         sigma ~ group
Data: df.variance (Number of observations: 60)
Samples: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup samples = 4000
```

Population-Level Effects:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	-0.01	0.73	-1.41	1.51	1.01	1107	1072
sigma_Intercept	1.15	0.17	0.85	1.51	1.00	1991	1922
group5yo	5.18	0.77	3.60	6.65	1.00	1252	1327
groupadults	7.98	0.74	6.47	9.37	1.01	1110	1079
sigma_group5yo	-1.05	0.24	-1.51	-0.57	1.00	2249	2420
sigma_groupadults	-2.19	0.24	-2.66	-1.74	1.00	2171	2427

Samples were drawn using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

← on a log scale!

```
mean = c(0, 5, 8)
sd    = c(3, 1.5, 0.3)
```

pretty
good!

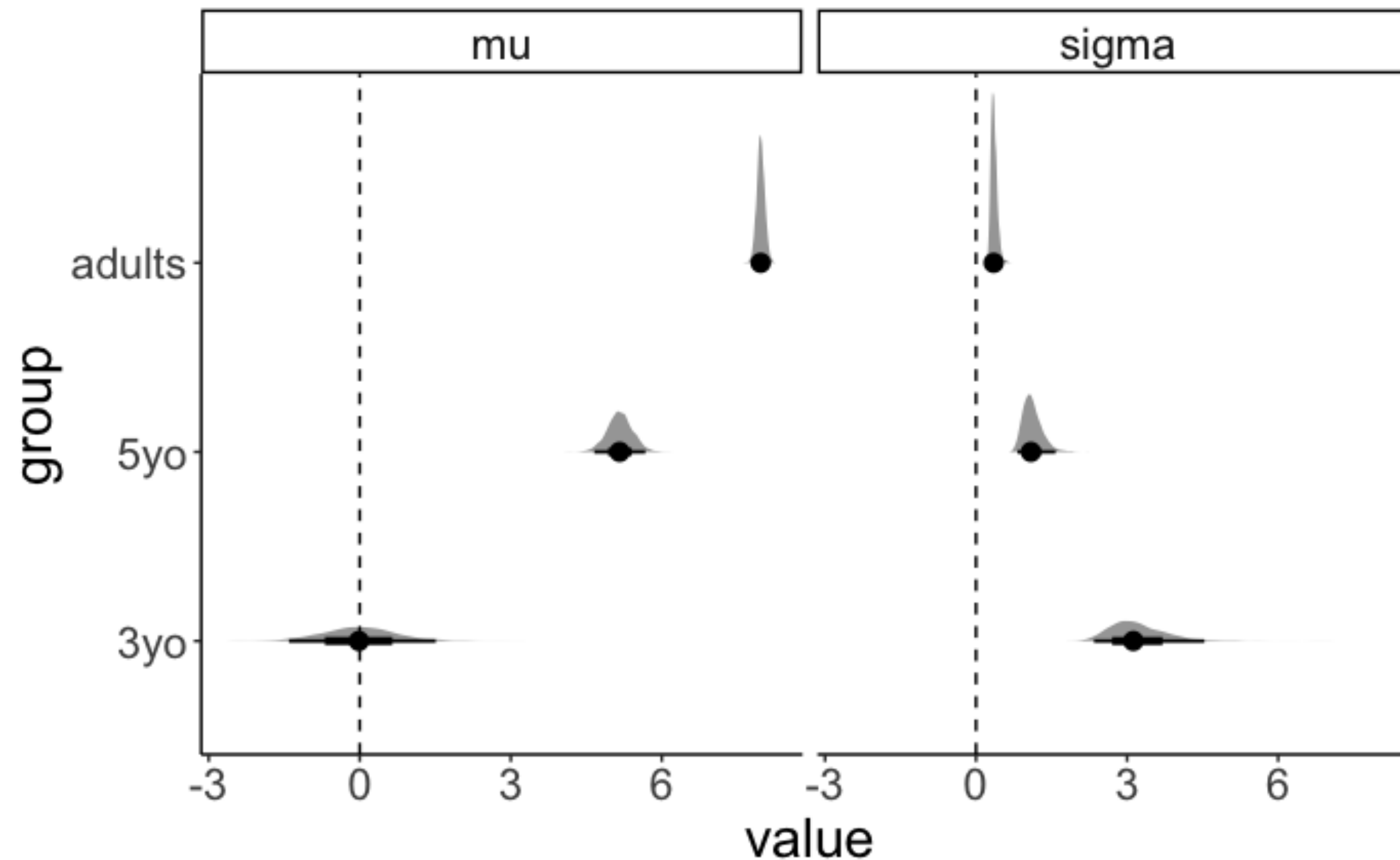
3 year olds $e^{1.15} = 3.16$

5 year olds $e^{1.15+(-1.05)} = 1.10$

adults $e^{1.15+(-2.19)} = 0.35$

Unequal variance aka heteroscedasticity

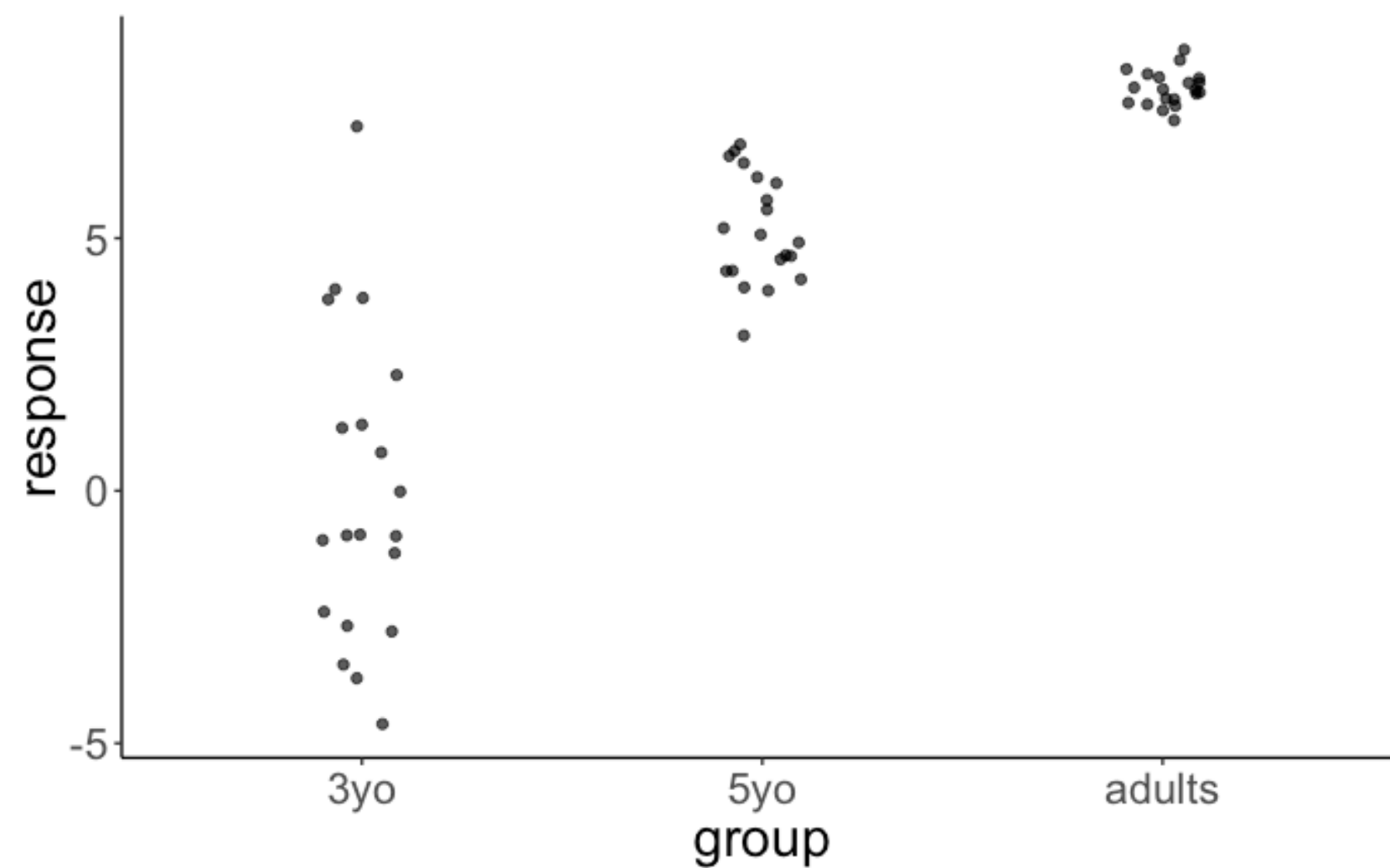
```
1 df.variance %>%
2   expand(group) %>%
3   add_fitted_draws(fit.brm1, dpar = TRUE) %>%
4   select(group, .row, .draw, posterior = .value, mu, sigma) %>%
5   pivot_longer(cols = c(mu, sigma),
6                 names_to = "index",
7                 values_to = "value") %>%
8   ggplot(aes(x = value, y = group)) +
9   geom_halfeyeh() +
10  geom_vline(xintercept = 0, linetype = "dashed") +
11  facet_grid(cols = vars(index))
```



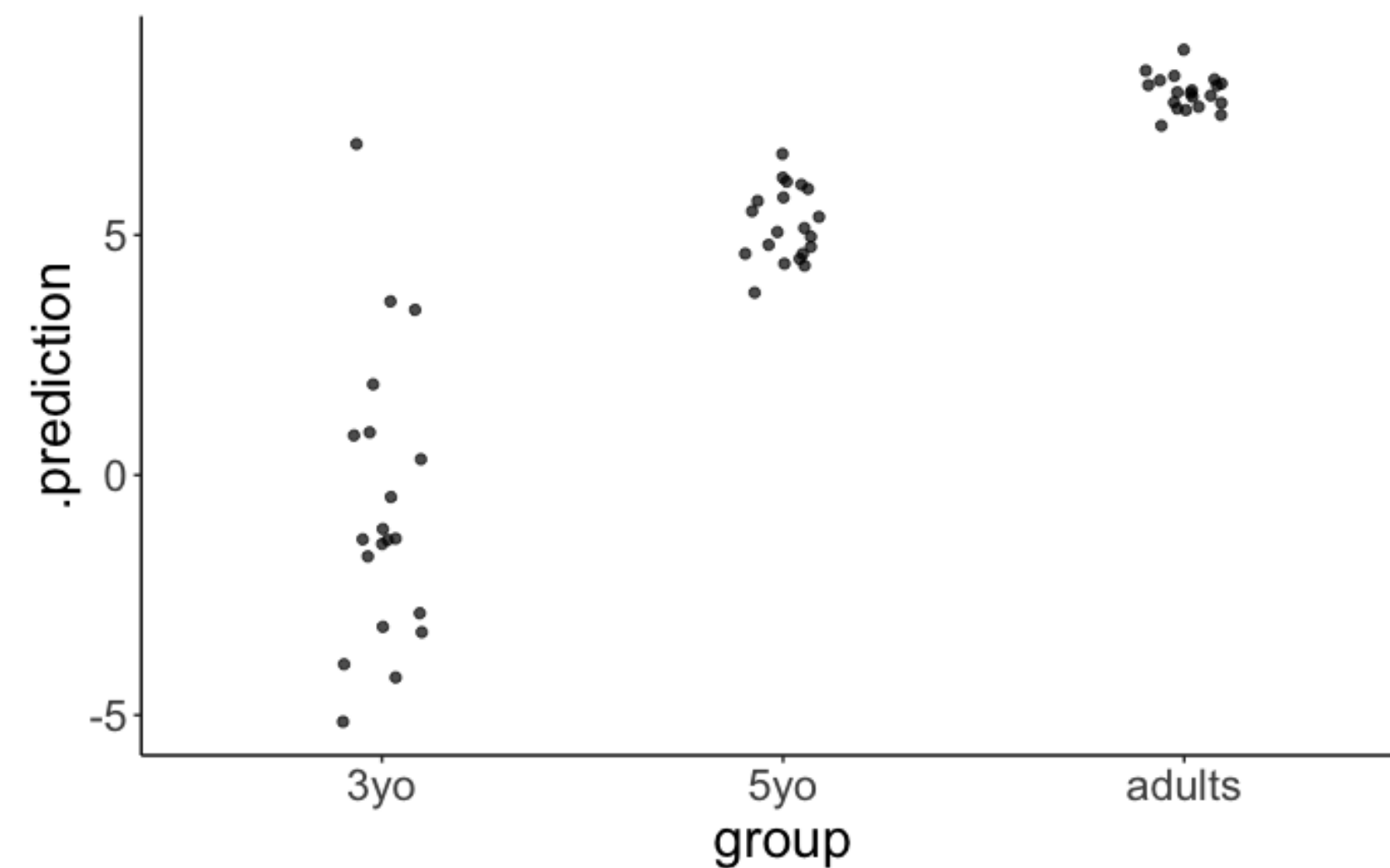
Unequal variance aka heteroscedasticity

```
1 df.variance %>%  
2   add_predicted_draws(model = fit.brm1,  
3                       n = 1) %>%  
4   ggplot(aes(x = group, y = .prediction)) +  
5   geom_jitter(height = 0,  
6               width = 0.1,  
7               alpha = 0.7)
```

original data



predicted data



these predictions look good!

Plan for today

- Quick recap
- Doing Bayesian data analysis **with BRMS**
 - Reporting results
- Some more examples
 - Sleep data
 - Titanic data
- Going beyond
 - Evidence for the null hypothesis
 - I want only positive coefficients
 - Dealing with unequal variance

Feedback

How was the pace of today's class?

much
too
slow

a little
too
slow

just
right

a little
too
fast

much
too
fast

How happy were you with today's class overall?



What did you like about today's class? What could be improved next time?

Thanks